



CRITICAL DESIGN REVIEW

OrbitMark

Final product-development baseline and UX/UI design authority

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Status: CONDITIONAL CDR APPROVAL - orbital-data and autonomous-delivery authority incorporated; mandatory gates remain
Classification: Confidential

Evidence-control statement

This document does not treat design choices, thresholds, forecasts, or unresolved legal questions as facts. Every substantive statement is classified as a verified fact, an approved design decision, a testable requirement, or an open item. External professional sign-off is still required where explicitly identified.

Document control

Field	Controlled value
Authority	This CDR controls product scope, architecture, data contracts, UX/UI requirements, verification gates and Claude Design handoff.
Supersedes	Technical and UX/UI decisions in the PDR and 11_DIAGRAMS.md where this CDR explicitly resolves a conflict.
Does not supersede	The business plan for commercial strategy and financial modelling; applicable law; platform policies; data-provider terms.
Source hierarchy	1) This CDR; 2) approved interface/data contracts; 3) approved design tokens and screen specifications; 4) diagrams; 5) PDR; 6) business plan.
Approval class	Conditional approval of the development plan. Production launch remains prohibited until the release gate is signed.
Review method	Structured multi-disciplinary adversarial review ("Council") across product, orbital mechanics, mobile architecture, UX/human factors, accessibility, security/privacy/legal, and QA/operations.
Current revision	v1.2 - retains binding Appendix G and adds Appendices H-J for autonomous delivery, skills/Council governance, Claude Design production and evidence automation for orbital-object ingestion, identity, propagation, working-set selection, pass prediction, validation, observability and UX obligations.

Classification legend

Mark	Meaning	Use in this CDR
F	Verified fact	Supported by an internal baseline document or a primary/official external source.
D	Design decision	A deliberate approved choice. It may be revised only through change control.
R	Requirement	A testable obligation with an acceptance method and owner.
T	Threshold	A project acceptance target; it is not presented as a universal industry fact.
O	Open item	Unresolved, blocked, or requiring external review. It cannot be silently assumed.

CDR result

APPROVE WITH CONDITIONS. The product is technically credible as an honest, model-based orbital-object companion centred on personal curation. The current baseline is not approved unchanged: the CDR mandates an OMM-first data model, a corrected propagation architecture, local-first catalog and pass computation, explicit non-AR parity, a dedicated ingest process, and measurable accessibility, sensor, privacy, claims and release gates.

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1. Purpose, scope and evidence protocol

The CDR converts the commercial thesis and preliminary design into a controlled, testable development baseline.

1.1 Purpose

The purpose of this Critical Design Review is to decide whether the proposed product design is sufficiently complete, internally consistent and verifiable to authorize detailed implementation. It is not a business plan, a marketing document or a certification. Its function is to establish the product contract that design, engineering, testing and launch review must follow.

The business plan defines the commercial thesis: OrbitMark is viable only as an annotation-first, education-framed product rather than another general satellite viewer [I-01]. The PDR established technical feasibility and identified the core constraints: public orbital elements and SGP4 make the orbital calculations tractable, while mobile pointing uncertainty and weak differentiation are the major design risks [I-02]. The diagrams supplied an initial architecture and interaction baseline [I-03]. This CDR accepts those foundations, corrects inconsistencies, and makes the resulting design testable.

1.2 In scope

- Consumer mobile product for iOS and Android, English-first, distributed through app stores after the web/PWA feasibility phase.
- Camera-backed sky view, manual/synthetic sky, object discovery, object detail, personal catalog, annotations, watch state, pass predictions and local alerts.
- Public orbital-element ingestion, provenance, storage, client caching, on-device propagation, optional catalog sync and education content.
- Information architecture, screen requirements, visual system, interaction states, accessibility, privacy, security, operations, analytics and release gates.
- Instructions and design-authority controls for Claude Design / Claude Code implementation.

1.3 Explicitly out of scope

- Conjunction assessment, collision probability, maneuver advice, operator-grade ephemerides, surveillance, or any other Space Situational Awareness service.
- Claims that the camera detects or visually identifies orbital debris; the product renders computed positions.
- Automatic identification of a single exact object from raw phone pointing when multiple candidates fall within uncertainty.
- Plate solving, social feeds, child accounts, classroom rosters, public user-generated content, advertising personalization, and institutional administration in MVP.
- Financial forecast approval. Commercial assumptions remain controlled by the business plan and require separate validation.

1.4 Evidence protocol

Rule	CDR application
Primary-source preference	Technical, standards, policy and platform claims use official documentation, normative standards or the primary source repository.
No false precision	Where sensor quality, visibility, data age or platform behaviour varies, the UI shows a state, range or explicit unknown rather than invented certainty.
No fact/decision mixing	Proposed thresholds are labelled T; architecture choices are labelled D; unresolved legal questions are O.
Traceability	Every mandatory requirement has an ID, rationale, verification method and release gate.
Conflict handling	If a prior diagram or narrative conflicts with this CDR, the CDR decision record controls.
Change control	A requirement, ADR or screen contract may change only through a recorded change note with impact on tests, data, UX and claims.

1.5 External sign-offs not represented as completed

Mandatory honesty boundary

The "Council" in this document is a structured adversarial review framework, not a claim that independent human lawyers, accessibility

auditors, orbital-dynamics specialists or security assessors have certified the product. Before production launch, those external reviews remain mandatory where listed.

2. Baseline review and Council determination

A red-team review of the three supplied baselines, with explicit resolution of contradictions.

2.1 Baseline documents

ID	Document	Role in CDR
I-01	OrbitMark Business Plan v1.0, June 2026	Commercial thesis, wedge, business constraints, market framing, kill criteria and verify-gated posture.
I-02	Orbital Debris AR App PDR v1	Technical feasibility, competitor and practitioner evidence, initial product characterization and risk framing.
I-03	11_DIAGRAMS.md	Initial system context, containers, sequences, data model, confidence state and milestone flow.

2.2 Council review lenses and determinations

Council lens	Adversarial question	Determination
Product & strategy	Does the design protect the annotation/curation wedge?	Approve only if saving, annotating, watching and returning to “your objects” is the primary loop. Catalog size and AR spectacle are secondary.
Orbital mechanics & data	Are data formats, propagation claims and identifiers future-safe?	Approve with OMM-first storage, large catalog identifiers, provenance, source-age display and independent numerical validation.
Mobile/AR architecture	Can a Capacitor client produce a trustworthy experience?	Approve as an orientation-based camera overlay, not world-anchored AR. Require a native sensor-plugin escape hatch and device field gate.
UX & human factors	Will users understand what is modelled, uncertain, stale or invisible?	Approve only with confidence states, candidate lists, data freshness, manual sky parity, permission-late onboarding and plain-language copy.
Accessibility	Can core jobs be completed without camera, motion, colour or fine pointing?	Approve only if the non-AR path is feature-equivalent and WCAG 2.2 AA/WCAG2ICT criteria are verified.
Security, privacy & legal	Is precise location or minor data unnecessarily exposed?	Approve with local-first location and pass computation, optional accounts, no child-directed/social MVP, provider-terms and store-policy review.
QA & operations	Can ingestion, computation, UI and release claims be observed and tested?	Approve with dedicated ingest execution, deterministic test vectors, telemetry, visual regression, rollback and release evidence packs.

2.3 Baseline consistency audit and mandatory resolutions

ID	Baseline issue	CDR resolution
C-01	PDR leaves React Native vs. Capacitor open; diagrams select React/Vite + Capacitor.	D-001: Capacitor-wrapped React/Vite is the MVP baseline because it preserves the proposed web stack. Native modules are an explicit escape hatch, not an unplanned rewrite.
C-02	Diagram labels “satellite.js WASM propagator.” satellite.js is a JavaScript/TypeScript library.	D-002: Correct to a TypeScript propagation Web Worker using satellite.js. WASM is not part of the approved baseline unless separately benchmarked.
C-03	Data model stores TLE line1/line2 while the product and providers recommend OMM.	D-003: OMM JSON is canonical; normalized fields and raw source payload are retained. TLE is compatibility input only. Catalog IDs use BIGINT/string-safe handling [E-01][E-02].
C-04	AR loop reads IndexedDB every ~200 ms.	D-004: IndexedDB is persistence, never the frame-loop data source. Parsed records are

ID	Baseline issue	CDR resolution
		held in memory inside a Web Worker; UI frames interpolate rendered positions.
C-05	Server precomputes passes and receives precise lat/lon.	D-005: MVP computes passes on-device and schedules local notifications. Precise location remains local by default; server pass computation is deferred.
C-06	In-process APScheduler appears inside the web deployment.	D-006: Ingestion runs in a dedicated process/cron with idempotency and locking. It is not started by Unicorn workers.
C-07	Diagram hard-codes an iOS pending-notification cap.	D-007: No fixed undocumented platform number is embedded in product logic. The client uses a rolling schedule, inspects pending notifications and reschedules on app activation.
C-08	"Decay outlook" risks implying a reliable prediction for every object.	D-008: Display actual decay date when sourced. Display an estimate only when a named source and uncertainty are available; otherwise show "Unknown."
C-09	Business plan includes education/minors as a channel, while social/shared catalogs appear later.	D-009: MVP is general-audience and educator-facilitated; no child accounts, roster data or public free-form sharing. Any children-directed scope triggers a separate compliance review.
C-10	AR can become the only discoverability path.	D-010: Manual Sky and Overhead list are first-class, not fallback-only. All core jobs must be possible without camera and motion permissions.

2.4 Council verdict

Conditional GO

The design is approved for implementation as a narrow, honest, annotation-first product. Approval is withdrawn if the product is repositioned as an exact camera detector, a general "largest catalog" tracker, an SSA tool, or a child-directed social platform without a new review.

3. Approved product definition and scope

The final product contract that design and engineering must implement.

3.1 Product statement

OrbitMark is a mobile sky companion that computes where catalogued artificial objects are expected to appear, explains what those objects are, and lets people save, annotate, watch and revisit the objects they care about. The camera view is an optional presentation layer over a modelled sky; it is not a visual detector. The persistent personal catalog is the core product.

3.2 Primary jobs to be done

Priority	User job	Required product outcome
P0	"Tell me what is likely in this part of the sky."	Show modelled candidates within an honest uncertainty region, ordered with visible provenance and data age.
P0	"Let me keep and follow this object."	Save locally without forced registration; add tags/notes; enable watch and next-pass alerts.
P0	"Show me what matters tonight."	Home and Passes surfaces prioritize watched objects and locally relevant passes.
P1	"Help me understand the object and why it matters."	Object detail distinguishes type/status, explains visibility limitations, and links every factual educational claim to a source.
P1	"Work when sensors, camera or network fail."	Manual Sky, Overhead list and cached data preserve the core experience with explicit degradation.
P2	"Back up or share my catalog."	Optional account sync and controlled share cards after the local loop is validated.

3.3 Personas controlled for design

Persona	Context	Needs	Design consequence
Curious newcomer	Arrives after seeing an object or a news story. Low orbital knowledge.	Immediate value, plain language, no forced permissions or account.	Home explains the next useful action. Technical detail is progressive.
Enthusiast curator	Tracks named spacecraft or debris and returns for passes.	Fast search, reliable watchlist, notes, tags, alerts and source depth.	Catalog and Passes are top-level destinations, not buried utilities.
Educator/facilitator	Uses the app with a group or classroom.	Credible facts, citations, manual mode, predictable presentation.	Education cards are sourced; no student accounts are required for MVP.
Accessibility-dependent user	May use screen reader, large text, switch/keyboard, reduced motion, or cannot hold a phone steadily.	Equivalent non-camera flow and robust controls.	Camera/motion are optional; no task depends on colour or drag alone.

3.4 Core product loop

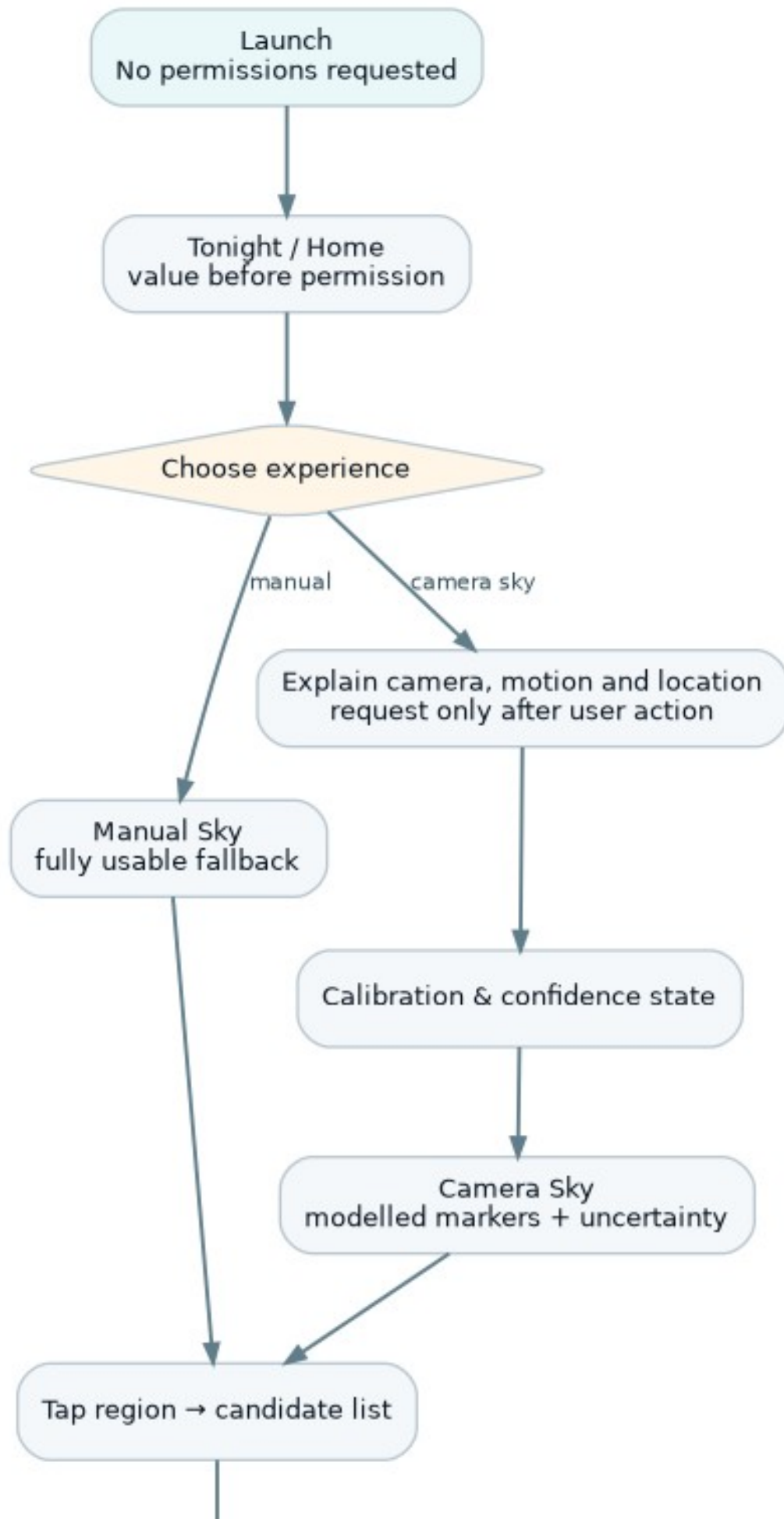


Figure 1 — Approved core user flow. The loop delivers value before requesting permissions and preserves a complete manual path.

3.5 MVP feature boundary

Capability	MVP	Later	Never implied
Manual/Synthetic Sky	Yes — first-class	Enhanced 3D globe	—
Camera-backed orientation overlay	Yes — conditional on device gate	Native tracking/plate solving if justified	Camera detection of debris
Candidate list from tap region	Yes	Smarter ranking	False single-object certainty
Object detail + provenance	Yes	Expanded histories	Unsourced decay prediction
Local catalog, notes, tags, watch	Yes	Cloud sync and shared collections	Public social feed in MVP
Pass predictions and local alerts	Yes, on-device	Optional server push	Guaranteed exact delivery time
Education cards	Yes, curated and cited	Facilitator mode	Unreviewed generated facts
Conjunction/collision tools	No	Separate thesis only	Operator-grade safety advice
Child accounts / classroom roster	No	Separate compliance program	Implicit collection of minor data

4. Requirements baseline

Mandatory functional and quality requirements, each testable and traceable.

4.1 Priority convention

P0 requirements are release-blocking. P1 requirements are required for public beta unless the Product Owner records a time-bounded waiver. P2 requirements are planned extensions and must not compromise the P0 architecture.

4.2 Functional requirements

ID	Pri.	Requirement	Verification
FR-001	P0	Launch without forced account, camera, motion, notification or precise-location permission.	UX inspection + clean-install test
FR-002	P0	Provide immediate value on Home using cached/global data or a clear setup state.	Usability + offline test
FR-003	P0	Offer Manual Sky and Overhead list as complete alternatives to Camera Sky.	Functional parity test
FR-004	P0	Request each permission only after a user action and after a plain-language primer states purpose and fallback.	Permission-flow test
FR-005	P0	Render modelled orbital-object positions from current cached elements and observer time/location.	Numerical validation
FR-006	P0	Display object-type shape, model status, data age and confidence state without relying on colour alone.	Visual/accessibility test
FR-007	P0	A tap on the sky returns a candidate list within the current uncertainty region; it does not silently choose one object.	Interaction test
FR-008	P0	Show object detail with name, catalog ID, object type/status, element epoch/source, position wording and visibility state.	Content contract test
FR-009	P0	Allow local save, unsave, note, tag and watch operations without registration.	Persistence test
FR-010	P0	Calculate upcoming passes for watched objects on-device using the same approved orbital-element set.	Cross-engine test
FR-011	P0	Schedule, inspect, cancel and roll forward local notifications; recover after permission or exact-alarm changes.	Platform test
FR-012	P0	Expose data freshness globally and per object; never silently use stale or failed data.	Failure injection test
FR-013	P0	Cache the latest valid data and preserve catalog operations offline.	Airplane-mode test
FR-014	P0	Provide clear calibration, low-confidence, denied-permission and unavailable-sensor states.	State-machine test
FR-015	P0	Make all core tasks possible with screen reader and without camera, motion, fine pointing, drag or colour discrimination.	Accessibility audit
FR-016	P1	Search and filter by object name, catalog ID, type, status, tags and watched state.	Functional test

ID	Pri.	Requirement	Verification
FR-017	P1	Show cited education content that distinguishes tracked objects from untracked modelled debris populations.	Editorial/source review
FR-018	P1	Allow export and deletion of local personal data; if sync exists, provide account deletion and server-side deletion.	Privacy test
FR-019	P1	Provide optional catalog sync without changing local-first behaviour.	Sync conflict test
FR-020	P1	Create share cards that state "modelled position" and omit precise user location.	Claims/privacy review
FR-021	P1	Provide an ingest/data-status surface for operational diagnosis.	Operations test
FR-022	P2	Provide 3D orbit visualization only after the core curation loop and performance budget pass.	Roadmap gate
FR-023	P2	Add plate-solving only after an independent feasibility and privacy review.	New CDR
FR-024	P0	Fetch orbital data only through documented provider interfaces, preserving response metadata, source class and provider timestamps.	Provider integration test
FR-025	P0	Validate, normalize and propagation-smoke-test every element record; invalid records enter quarantine and are excluded from publication.	Schema/range/failure injection suite
FR-026	P0	Store raw source payloads and normalized element sets immutably; publication selects a current record without overwriting history.	Database immutability test
FR-027	P0	Exclude reused analyst-range and temporary launch-nominal identities from the MVP personal catalog and public package unless a lineage-safe identity contract is approved.	Identity fixture test
FR-028	P0	Publish only immutable, checksummed packages with a manifest and atomically move the last-known-good pointer after complete validation.	Package atomicity test
FR-029	P0	Verify package checksum and compatibility before client activation; preserve the previous package when verification or parsing fails.	Client rollback test
FR-030	P0	Use the approved OMM -> SatRec -> SGP4/SDP4 -> Earth-fixed -> topocentric look-angle pipeline with explicit TEME-oriented variable naming.	Numerical pipeline test
FR-031	P0	Select a bounded, prioritised working set outside the animation loop; watched/selected objects receive highest computation priority.	Worker scheduling/performance test
FR-032	P0	Compute rise, culmination and set for watched objects with bracketed root finding and bounded maximum-elevation search using the same element set and observer model.	Pass-event cross-engine test
FR-033	P0	Represent geometric elevation,	Domain/content contract test

ID	Pri.	Requirement	Verification
		sunlight state, brightness knowledge, sky darkness and model freshness as separate fields; do not collapse them into a false visibility claim.	
FR-034	P0	Expose structured ingest, package, propagation, clock and sensor failure states to operations and map user-relevant states to Data Status and object detail.	Failure-state mapping test
FR-035	P1	Provide a developer/operations diagnostic export containing package version, source, element epoch, calculation time, frame convention and non-sensitive error codes.	Supportability test

4.3 Non-functional requirements

ID	Pri.	Requirement	Verification
NFR-001	P0	Orbital computations are deterministic for fixed OMM/time/location input and validated against an independent SGP4 implementation.	Automated reference-vector suite
NFR-002	P0	The UI never promises sensor accuracy that the client cannot measure. Confidence labels are based on observable quality signals or are explicitly qualitative.	Code/content review
NFR-003	P0	Sky rendering remains responsive on the supported baseline device matrix using worker-based computation and bounded object sets.	Performance profile
NFR-004	P0	The client does not read IndexedDB or perform full-catalog propagation in the animation frame.	Architecture/code review
NFR-005	P0	Catalog IDs support current and future identifiers beyond five digits; no TLE-width assumption exists in storage or UI.	Schema/property tests
NFR-006	P0	Ingestion is idempotent, provider-friendly, rate-limited, retryable and independently observable.	Integration/failure test
NFR-007	P0	Precise location remains on-device by default and is not included in analytics, logs, share cards or optional sync records.	Privacy data-flow audit
NFR-008	P0	Meet WCAG 2.2 Level AA for web surfaces and apply WCAG2ICT guidance to mobile software; use native accessibility APIs.	Independent audit + automated checks
NFR-009	P0	Apply OWASP MASVS controls appropriate to a consumer app and OWASP ASVS controls to the backend.	Security checklist + penetration test
NFR-010	P0	All factual educational and product claims have a source ID and review date.	Claims registry audit
NFR-011	P0	Telemetry is first-party, minimal, pseudonymous and disabled for data categories not required to operate the product.	Data inventory review
NFR-012	P1	Architecture is localization-ready and avoids text embedded in images or fixed-width labels.	Pseudo-localization test

ID	Pri.	Requirement	Verification
NFR-013	P1	The app supports dynamic text, screen rotation where safe, reduced motion and light/dark or night-appropriate contrast.	Platform accessibility test
NFR-014	P1	A failed data refresh never deletes the last known-good element set.	Failure injection test
NFR-015	P1	All schema and API changes are versioned and backwards compatible within the supported client window.	Contract test
NFR-016	P1	Production jobs do not run implicitly in each web worker process.	Deployment inspection
NFR-017	P1	Release artifacts include SBOM/dependency scan, privacy manifest/data-safety declarations and reproducible version metadata.	Release evidence pack
NFR-018	P0	All orbital timestamps are parsed and stored as UTC instants; timezone-free or ambiguous timestamps are rejected rather than guessed.	Time parsing property tests
NFR-019	P0	Internal orbital state names state the frame and units (for example position_teme_km); generic x/y/z or eci names are not accepted at module boundaries.	Static contract review
NFR-020	P0	Provider credentials, if Space-Track is used, remain server-side and never ship in the client package or logs.	Secret scanning + architecture test
NFR-021	P0	Package publication, current-pointer change and client activation are atomic and recoverable; no partial package is observable as current.	Transaction/failure injection test
NFR-022	P0	Pass calculations and sky positions are reproducible for fixed package, UTC time and observer input across supported client engines within approved tolerance.	Golden fixture suite
NFR-023	P0	The app detects material device clock skew and labels calculations/alerts as degraded rather than silently presenting false timing precision.	Clock-skew simulation
NFR-024	P1	Ingest polling obeys provider cadence, conditional requests, stop-on-error and exponential backoff; repeated 403/429/5xx responses trigger operator action.	Provider policy test
NFR-025	P1	The worker maintains an in-memory parsed set and publishes bounded typed messages; IndexedDB access, JSON parsing and catalog-wide sorting are absent from the render frame.	Profile + code inspection
NFR-026	P1	Numerical fixtures include near-Earth, deep-space, high-eccentricity, decayed, malformed, large-ID and stale-element cases.	Fixture coverage report

4.4 Key acceptance thresholds

ID	Area	Project threshold
T-001	Math pipeline	For fixed OMM/time/location vectors, angular output differs from the independent reference by no more than the project tolerance approved during M0; initial target $\leq 0.1^\circ$.

ID	Area	Project threshold
T-002	Camera Sky field quality	On the approved device matrix, the displayed uncertainty region contains the known bright reference object in at least 80% of calibrated trials. Results are reported by device, environment and orientation.
T-003	Core task success	At least 85% unassisted completion for: open Manual Sky, identify candidates, save object, enable watch, find next pass. No critical misunderstanding of “modelled position.”
T-004	Accessibility	Zero release-blocking WCAG 2.2 AA / platform accessibility defects; all core tasks pass VoiceOver and TalkBack manual testing.
T-005	Performance	Camera/Manual Sky maintains a visually stable interaction target on supported devices; the specific frame-time budget is fixed after M0 profiling and enforced in CI.
T-006	Data freshness	The app exposes element epoch and fetch state. Alert generation is suppressed or labelled when data exceeds the empirically approved maximum age.
T-007	Crash quality	No known P0/P1 crash; beta crash-free session target is set after instrumentation and reviewed at release gate.
T-008	Ingest completeness	A candidate package is not published if the manifest count/checksum disagrees with stored records, required validation jobs fail, or the invalid-record rate exceeds the M0-approved guardrail.
T-009	Provider conduct	No source is polled more frequently than its documented update cadence; on 403/429 the worker stops the run and does not retry aggressively.
T-010	Pass timing	Rise/set/maximum event times match the independent reference implementation within the M0-approved tolerance; initial target is ≤ 2 seconds for controlled fixtures.
T-011	Clock integrity	A device/server time discrepancy above the M0-approved threshold produces a visible degraded state and prevents high-confidence alert wording; initial evaluation threshold is 30 seconds.
T-012	Working-set budget	The active propagation set and cadence satisfy the supported-device CPU, memory, frame-time and battery budget fixed at M0; the full catalog is never propagated per visual frame.
T-013	Identity safety	Zero personal annotations or watch records are attached to a different physical object after catalog-number reuse in analyst/temporary identity fixtures.
T-014	Package rollback	Every simulated fetch, parse, validation, write or activation failure leaves a readable last-known-good package and a diagnosable state.

Threshold discipline

The values above are release criteria chosen for this product. They are not represented as universal scientific or industry facts. M0 may tighten or relax a threshold only with recorded evidence and Council approval.

5. Approved system architecture

A local-first mobile architecture that separates persistence, propagation, presentation and operations.

5.1 Architecture overview

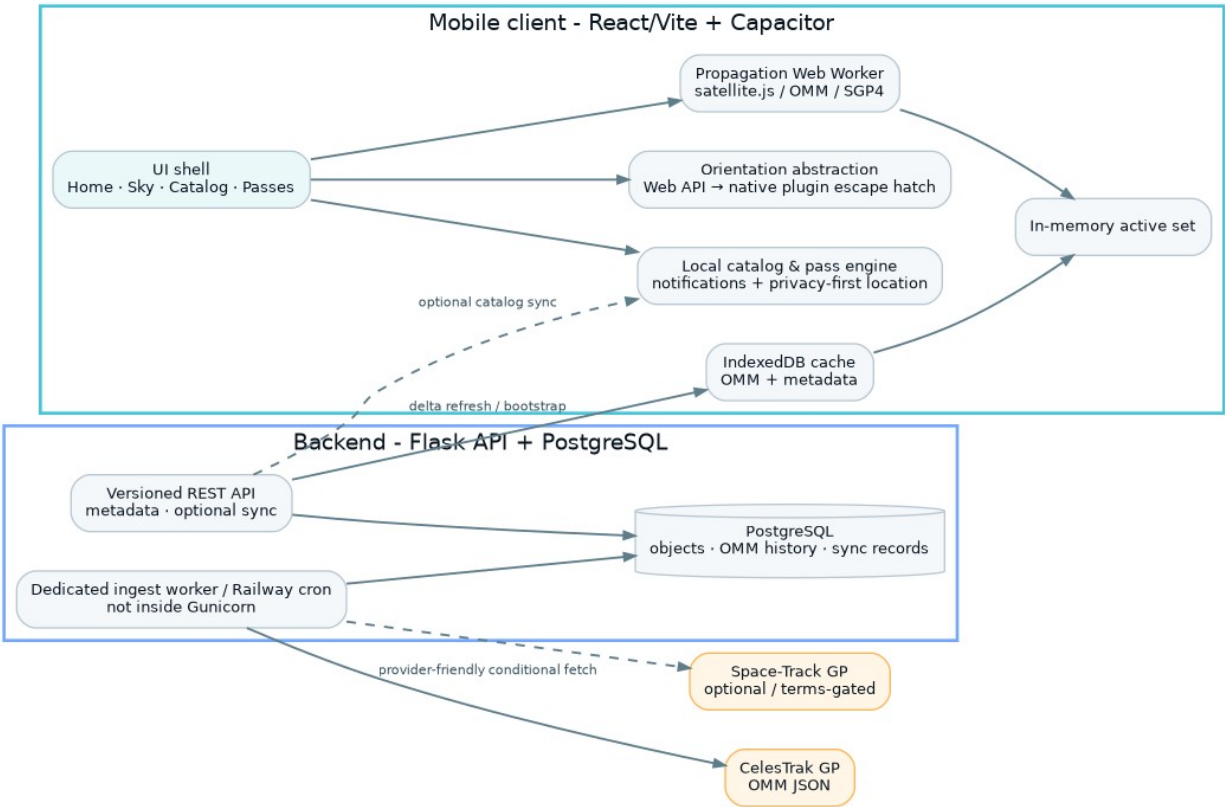


Figure 2 — Approved container architecture. Major changes from the supplied diagram are highlighted in the baseline consistency audit.

5.2 Architecture principles

- **Local-first core:** Camera/Manual Sky, catalog operations, pass computation and precise location work without a server round trip.
- **OMM-first data:** The canonical orbital payload is OMM-compatible JSON with provenance and large identifiers [E-01] [E-02].
- **Worker isolation:** Parsing and propagation run in a Web Worker. The main thread renders bounded results and handles interaction.
- **Persistence is not the render loop:** IndexedDB stores the last known-good package; active parsed records live in memory.
- **Optional account:** Authentication is introduced only for sync/backup, not for initial exploration or local curation.
- **Operational separation:** Web serving and scheduled ingestion are separate processes with explicit locking and observability.
- **Native escape hatch:** The orientation abstraction can use Capacitor Motion first and a thin native plugin if the field gate fails.

5.3 Architecture decision records

ADR	Status	Topic	Decision	Rationale
ADR-001	Accepted	Client baseline	React + TypeScript + Vite, wrapped with Capacitor for iOS/Android after the browser/device spike.	Preserves the established stack and supports incremental native plugins.
ADR-002	Accepted	AR definition	Camera passthrough plus orientation-based overlay. No world anchors or	Matches the actual sensor/data capability and avoids false claims.

ADR	Status	Topic	Decision	Rationale
			camera-object recognition in MVP.	
ADR-003	Conditional	Orientation source	Begin with standards-based/Capacitor motion APIs; escalate to native iOS/Android heading APIs when confidence quality or device consistency is insufficient.	Web orientation permission and absolute heading support vary; the abstraction prevents lock-in [E-14][E-15].
ADR-004	Accepted	Orbit library	satellite.js TypeScript in a Web Worker, validated against python-sgp4 or reference vectors.	Primary library supports OMM/TLE, SGP4/SDP4 and coordinate transforms [E-03].
ADR-005	Accepted	Data format	OMM-compatible JSON is canonical. Raw payload + normalized fields are stored.	TLE format has identifier limitations; providers recommend OMM migration [E-01][E-02].
ADR-006	Accepted	Pass engine	On-device pass generation for watched objects and local notifications.	Reduces location collection, server cost and online dependency.
ADR-007	Accepted	Catalog	Local database first; optional versioned sync later.	Removes registration friction and protects the core wedge offline.
ADR-008	Accepted	Backend	Flask versioned API + PostgreSQL for orbital metadata, ingestion provenance and optional sync.	Fits founder capability and existing operating model.
ADR-009	Accepted	Scheduler	Dedicated worker/Railway cron; no in-process scheduler in Unicorn.	Avoids duplicate jobs and lifecycle ambiguity.
ADR-010	Accepted	Education content	Curated source-controlled content; no unreviewed generative claims in production.	Trust is a product requirement.
ADR-011	Accepted	Audience	General audience; educator-facilitated use; no child accounts or public social features in MVP.	Avoids silently entering children/social policy scope [E-12][E-13].
ADR-012	Accepted	Accessibility	Manual Sky/Overhead is an equivalent product path, not a reduced fallback.	Core jobs cannot depend on camera, motion or visual pointing.
ADR-013	Accepted	Source precedence	CelesTrak OMM-compatible JSON is the default public MVP feed. Space-Track is an authenticated secondary/history source only after provider/legal review. SupGP remains a separately labelled source lane and is never silently mixed.	Preserves simple public ingestion, explicit provenance and provider compliance [E-01][E-02].
ADR-014	Accepted	Object identity	A stable internal object_key is separate from provider catalog identifiers. Analyst-range 80000–89999 and temporary launch-nominal identities are excluded from user-visible MVP persistence unless lineage-safe mapping is approved.	Space-Track states analyst numbers are reusable; a number alone is not durable identity [E-02].
ADR-015	Accepted	Frame contract	The SGP4 result is represented internally as position_teme_km and velocity_teme_km_s. The satellite.js ECI-labelled API may be used only behind an adapter that documents its SGP4/TEME semantics.	Prevents ambiguous frame mixing and future integration errors [E-03] [E-20][E-22].
ADR-016	Accepted	Package model	Server publication produces immutable	Provides rollback, auditability and

ADR	Status	Topic	Decision	Rationale
			versioned packages and manifests; the current pointer moves atomically after full validation. Clients verify checksum and compatibility before activation.	deterministic calculations.
ADR-017	Accepted	Working-set scheduler	Three priority bands are used: selected/watched, above-horizon/camera candidates, and background eligibility. Cadences and counts are profiling outputs, not hard-coded assumptions.	Controls CPU/battery while preserving user value.
ADR-018	Accepted	Pass solver	Passes use adaptive coarse sampling, sign-change bracketing, bisection for rise/set and a bounded scalar search for maximum elevation.	Produces deterministic event boundaries without depending on a server or derivative availability.
ADR-019	Accepted	Time authority	UTC instants control propagation. Device time is checked against a trusted response timestamp when online; material skew creates a degraded state.	Timing error directly affects sky position and notifications.
ADR-020	Accepted	Visibility domain	Geometric position, sunlight/shadow, magnitude/brightness knowledge, observer darkness and source freshness remain separate model fields.	Prevents “above horizon” from being presented as “visible.”

5.4 Client components

Component	Responsibility
AppShell	Routing, navigation, global data/privacy state, deep links and error boundary.
Home/Tonight	Next watched pass, overhead objects, freshness state and primary “Open Sky” action.
SkyView	Camera or synthetic background, marker projection, reticle, confidence state, filters and tap-region interaction.
OrientationService	Permission, screen-orientation transform, heading/elevation/roll normalization, smoothing, quality state and manual offset.
OrbitWorker	OMM parsing, satrec cache, subset selection, propagation, shadow/visibility support and pass calculations.
ElementStore	IndexedDB packages, version metadata, checksum, last-known-good and migration.
CatalogStore	Local objects, notes, tags, watch status, conflict metadata and export.
NotificationService	Permission, rolling local schedule, cancellation, pending-state inspection and app-activation reschedule.
AccessibilityAdapter	Accessible names, announcements, list alternative for sky content, focus management and reduced motion.
Telemetry	Minimal event capture with no precise location or note content.

5.5 Backend components

Component	Responsibility
Public metadata API	Versioned object metadata and compressed element packages/deltas, with ETag and source provenance.
Optional sync API	Authenticated catalog backup, tombstones, version conflicts, export and deletion.

Component	Responsibility
Ingest worker	Conditional provider fetch, schema validation, deduplication, checksum, provenance, publish and alerting.
Database	Objects, current/historical element sets, ingest runs, source state and optional user sync records.
Operations surface	Freshness, last success, failed records, provider response, package version and client compatibility.

5.6 Deployment requirements

- Production web/API and ingestion use independent entry points. An ingest lock and unique source/version key make runs idempotent.
- Every deploy has a version, commit, schema revision and rollback target. Database migrations are forward-tested and backup-aware.
- Client packages are immutable and checksummed. A failed or partial update never replaces the last valid package.
- Provider polling respects documented cadence and backoff. CelesTrak explicitly warns against redundant requests and provides OMM/JSON paths [E-01].
- Feature flags control Camera Sky, optional sync, education content and experimental ranking independently.

6. Orbital data, computation and uncertainty

The physics and data contract that prevent misleading output.

6.1 Authoritative data position

Celestrak and Space-Track publish general perturbations element data in OMM-compatible formats and support identifiers beyond the legacy five-digit TLE range [E-01][E-02]. `satellite.js` supports TLE or OMM inputs and implements the SGP4/SDP4 calculations and coordinate transforms required for a browser/client implementation [E-03]. These facts support a technically feasible consumer visualization; they do not make the phone's camera or compass an object detector.

6.2 Approved data lifecycle



Figure 3 — Orbital data lifecycle. Persistence, parsing, selection, propagation and rendering are separated.

6.3 OMM-first schema rules

Rule	Mandatory implementation
Catalog identifier	Database BIGINT; API JSON string or safe integer contract; UI never truncates or assumes five digits.
Raw preservation	Store the source OMM-compatible payload, source URL/class, fetch time, checksum and parser version.
Normalized fields	Store epoch, mean motion, eccentricity, inclination, RAAN, argument of pericenter, mean anomaly, BSTAR and classification fields needed by the selected library.
Provenance	Every displayed position is traceable to source, element epoch, ingest run and client package version.
History	Retain sufficient prior sets for audit/debugging; retention period is an operational decision documented before production.
Invalid record	Quarantine invalid or unparseable records; do not publish the entire package as valid if integrity checks fail.
Decay	Actual/source-reported decay is separate from any estimate. No generic "decay outlook" field without source and uncertainty.

6.4 Computation pipeline

- 27. Load the last valid OMM package from IndexedDB during startup and fetch a version/delta in the background.
- 28. Parse OMM records into `satellite.js` records inside a Web Worker; keep the active set in worker memory.
- 29. At a controlled cadence, select objects relevant to the observer horizon, user filters and watched catalog. Do not propagate the full catalog every visual frame.
- 30. Propagate selected objects for the current timestamp; convert from ECI to Earth-fixed/topocentric altitude and azimuth using the approved observer coordinates and time source.
- 31. Project altitude/azimuth through the orientation transform into screen coordinates; interpolate between computation ticks for visual stability.
- 32. Render only objects that satisfy horizon, clutter and confidence rules. Candidate search uses angular separation, not screen-pixel proximity alone.
- 33. Attach source age, object type, position wording and visibility status to the rendered model.

6.5 Visibility wording

State	Permitted copy	Prohibited implication
Visible likely	"May be visible under current conditions."	Guaranteed naked-eye visibility.
Visibility unknown	"Visibility is unknown."	Treating missing magnitude as invisible.
Not expected visible	"Not expected to be visible to the naked eye."	The camera has detected it.
Sunlit/model support	"Model indicates the object is sunlit."	Photometric certainty without atmospheric/brightness data.
Debris	"Catalogued debris / rocket body at a modelled position."	"You are seeing the debris through the camera."

6.6 Orientation and uncertainty model

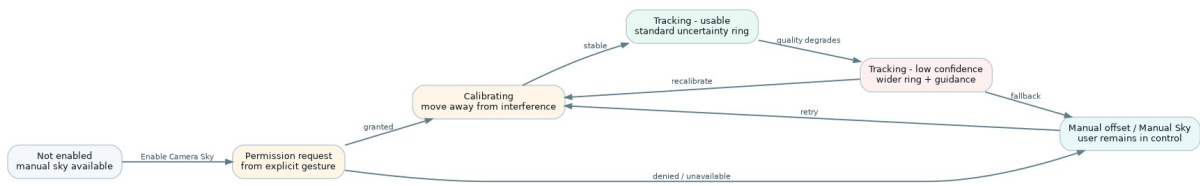


Figure 4 — Approved confidence and fallback state machine. The user always retains a manual path.

The orientation service must normalize device and screen coordinate frames, apply smoothing, preserve manual offset and expose a quality state. The Device Orientation specification defines orientation relative to an implementation-defined reference frame and distinguishes relative from absolute orientation [E-15]. Capacitor’s Motion API uses these web capabilities and requires permission handling on relevant platforms [E-14]. Therefore, “heading accuracy in degrees” is not assumed to exist consistently in the web layer.

Confidence state	Trigger evidence	UI behaviour
Not enabled	No permission or user has not asked for Camera Sky.	Manual Sky works; show a single “Enable Camera Sky” action with no nagging.
Calibrating	New session, unstable heading, orientation discontinuity or manual reset.	Show short instruction; suppress false precision; keep markers secondary.
Usable	Stable recent signal and calibration within approved test behaviour.	Show standard uncertainty ring and normal candidate interaction.
Low confidence	Signal instability, implausible jump, magnetic interference indication or failed field condition.	Widen/qualify the uncertainty region; explain corrective action; offer Manual Sky.
Manual	Permission denied/unavailable or user preference.	Drag/pan sky; preserve search, candidates, detail, save and passes.

6.7 Claims gate

Non-negotiable product language

Every surface that could be mistaken for visual detection must communicate “modelled position” in the interaction, detail view or persistent help. Marketing, store screenshots and share cards may not use “detect,” “scan,” “identify the exact fragment,” “live surveillance,” or equivalent wording unless a future system actually performs and validates that function.

6.8 Binding engineering authority

Appendix G is a normative part of this CDR. It controls source precedence, object identity, analyst-object handling, ingest state transitions, OMM normalization, immutable storage, package publication, UTC/time handling, SGP4 frame conventions, working-set selection, pass prediction, failure codes, test vectors, observability and the UX states that expose data quality. Where Appendix G is more specific than Sections 4–6 or the preliminary diagrams, Appendix G wins. No development team may substitute a different frame convention, identity rule, pass solver or data-freshness policy without a recorded ADR and CDR change approval.

7. UX/UI design authority

The controlling design chapter for Claude Design, engineering and design QA.

7.1 Standards and skill basis

The UX process follows human-centred design principles across the product lifecycle [E-10]. Accessibility requirements use WCAG 2.2 and its four principles, with WCAG2ICT guidance for applying Level A/AA criteria to non-web software [E-04] [E-05]. Interactive web patterns follow WAI-ARIA Authoring Practices where native semantics are insufficient [E-06]. Platform design must respect Apple Human Interface Guidelines for augmented reality and accessibility [E-07][E-08], and Material 3 accessibility guidance on Android [E-09].

For AI-assisted design, the approved primary skill is Anthropic's official frontend-design skill, which emphasizes deliberate subject-specific visual direction, typography, content, restraint, responsive design, visible keyboard focus and reduced motion [E-18]. The implementation workflow uses Claude Code skills and verification capabilities as documented by Anthropic [E-19], supplemented by app testing. Community design skills may generate alternatives but never override this CDR, platform guidelines or accessibility tests.

7.2 Product experience principles

ID	Principle	Design consequence
P-UX-01	Truth is visible	Model status, source age, confidence and visibility uncertainty are part of the primary interface, not buried legal text.
P-UX-02	Value before permission	Users see Tonight/Manual Sky before camera, motion, location, notification or account requests.
P-UX-03	One clear primary action	Each screen has one dominant next action. Secondary actions are quiet and contextual.
P-UX-04	Curation over spectacle	The app is remembered for “my objects,” not for a decorative AR effect.
P-UX-05	Progressive disclosure	Newcomers see plain language; enthusiasts can expand orbital/source detail.
P-UX-06	Graceful degradation	Offline, stale data, denied permissions and weak sensors produce a useful controlled state.
P-UX-07	Equivalent non-AR path	Manual Sky and list views complete the same jobs as Camera Sky.
P-UX-08	Calm night instrument	The visual system is dark, legible and disciplined; it avoids generic neon sci-fi ornament and excessive animation.
P-UX-09	Recognition over recall	Object types, watch state, data freshness and navigation remain visible and consistent.
P-UX-10	No inaccessible shortcuts	No colour-only meaning, drag-only action, tiny target, motion-only instruction or unlabelled canvas control.

7.3 Information architecture

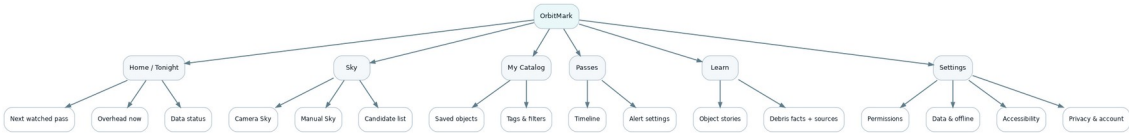


Figure 5 — Approved information architecture. Six top-level destinations preserve the curation loop and a complete non-AR path.

7.4 Navigation model

Surface	Navigation rule
Mobile app	Bottom navigation: Tonight, Sky, Catalog, Passes. Learn and Settings are reachable from top-level actions/menu. Four primary tabs only.
Sky	Camera/Manual mode is a segmented choice within one destination; switching modes preserves time, filters, selection and observer context.

Surface	Navigation rule
Object detail	Opens as a full-height sheet or page with a stable back path. Save/Watch remains visible without obscuring source/freshness.
Candidate list	Bottom sheet with accessible list semantics; focus moves to sheet heading and returns to the originating sky control on close.
Deep links	Pass alerts open the corresponding pass/object context, not a generic home screen.
Desktop/PWA	Responsive shell may use side navigation, but information hierarchy and labels remain identical.

7.5 Visual direction

Approved design thesis

OrbitMark should feel like a trustworthy field instrument used at night: precise typography, restrained orbital geometry, clear status language and one memorable signature — the uncertainty ring as both a visual motif and an honesty mechanism. It must not look like a game HUD, military tracker or generic purple-gradient AI dashboard.

Dimension	Approved direction	Avoid
Palette	Deep navy surfaces, near-white text, orbital cyan for primary action, amber for caution, blue/green supporting states.	Pure black everywhere, low-contrast grey, decorative rainbow coding.
Typography	A characterful but legible display face for product/section titles; highly readable sans-serif for body/data; tabular numerals for time/angles.	Multiple novelty fonts, condensed body text, all-caps paragraphs.
Shape	Moderate 12–16 px radii; thin orbital lines; clear containment.	Every element as a pill; excessive glassmorphism; faux 3D.
Iconography	Simple line icons plus geometric object-type shapes; labels always available.	Unlabelled astronomy glyphs or colour-only type coding.
Motion	Short purposeful transitions, sheet continuity, marker smoothing. Respect reduced-motion settings.	Parallax, starfield loops, constant pulsing or motion that implies accuracy.
Content density	Progressive detail; core facts visible, advanced orbit values collapsible.	Dense analyst tables on primary consumer screens.

7.6 Core design tokens

Token	Value	Use / contrast note
color.bg.canvas	#07131F	Main night canvas.
color.bg.surface	#0E2233	Primary app surface.
color.bg.panel	#132C40	Cards, sheets and controls.
color.text.primary	#F4F7FA	Primary text; high contrast on approved dark surfaces.
color.text.secondary	#A9BBC9	Secondary text; remains AA contrast on approved dark surfaces.
color.action.primary	#43C6D9	Primary action and active markers; use dark text on filled button.
color.focus	#74A7FF	Visible focus ring, not status colour.
color.warning	#FFB454	Low confidence, aging data and caution.
color.danger	#FF6B6B	Errors/destructive actions only.
color.success	#53D49A	Fresh/complete state; never sole indicator.
space.unit	4 px	All spacing uses a 4 px base; primary rhythm 8/12/16/24/32.
radius.control	12 px	Buttons, chips, fields.
radius.panel	16 px	Cards and sheets; avoid universal pills.
target.minimum	44 × 44 CSS px or platform equivalent	Project baseline; verify against WCAG/platform guidance.

Token	Value	Use / contrast note
motion.fast	150 ms	Micro state change.
motion.standard	220 ms	Sheet/navigation transition; disable/reduce when requested.

7.7 Type system

Role	Specification	Usage
Display	Candidate: Space Grotesk Semibold, 28–32 px	Product wordmark and rare hero/section title.
Title	System/IBM Plex Sans Semibold, 20–24 px	Screen titles and object names.
Body	System/IBM Plex Sans Regular, 16 px minimum target	Instructions, descriptions, education content.
Data	IBM Plex Sans/Mono with tabular numerals, 14–16 px	Time, angle, NORAD ID and source age.
Caption	13–14 px, high-contrast secondary	Metadata only; never critical instructions below readable size.

Font families are design candidates, not immutable facts. Before bundling, confirm licensing, platform rendering, dynamic type behaviour and performance. System-font fallback is mandatory.

7.8 Object-marker system

Object class	Shape	Colour role	Required label
Active payload	Circle	Cyan	Name or abbreviation + “active payload” in candidate/detail.
Inactive payload	Square	Muted blue	“Inactive payload.”
Rocket body	Diamond	Amber	“Rocket body.”
Debris	Triangle	Blue	“Catalogued debris.”
Watched object	Outer halo/bookmark state	Independent of type colour	“Watched” accessible state.
Uncertainty	Dashed/soft ring	Matches marker or neutral	Accessible description includes qualitative confidence.
Cluster	Count badge + list action	Neutral	“N modelled objects in this region.”

7.9 Component contract

Component	Contract
Primary button	One per view; verb-first label; 44 px target; loading, disabled and focus states; no icon-only primary action.
Status chip	Short state + icon/text. Never colour-only. Examples: “Data 4h old,” “Low confidence,” “Offline.”
Object row	Type shape, name, concise metadata, watch state, accessible combined label and predictable disclosure action.
Object card	Name/ID, type/status, source age, next pass, visibility, save/watch action and expandable source/orbit details.
Candidate sheet	Heading with candidate count, ordered list, explanation that positions are modelled, close/focus contract.
Freshness banner	Appears when refresh fails, data is aging/stale or source status is unknown. Provides next action and preserves last valid data.
Permission primer	Explains benefit, requested resource, local processing where applicable, fallback and “Not now.”
Calibration prompt	One instruction at a time; never blocks Manual Sky; no indefinite spinner.
Empty state	States why it is empty and the one action that resolves it.
Error state	Names the failed operation, preserves user data, offers retry/fallback and has a diagnostic code only in secondary detail.
Toast/announcement	Confirms reversible local action; screen reader live-region equivalent; never sole error channel.

Component	Contract
Bottom sheet/dialog	Platform-appropriate focus trap/return, escape/back behaviour and accessible title.

7.10 Permission strategy

Permission	When requested	Primer promise	Fallback
Camera	Only after “Enable Camera Sky.”	Used as a background for modelled markers; image is not uploaded by default.	Manual Sky.
Motion/orientation	Same explicit Camera Sky action or a distinct step if platform requires.	Aligns the modelled sky; may be inaccurate near interference.	Manual Sky + manual offset.
Approx./precise location	When user asks for local overhead/passess or chooses “Use current location.”	Used locally for sky/pass calculations; precise coordinates stay on device by default.	Search/select city or manual coordinates stored locally.
Notifications	After the user watches an object and sees the next pass value.	Alerts for selected watched objects; timing may depend on OS settings.	In-app Passes timeline.
Account/sign-in	Only when enabling sync/backup or cross-device restore.	Backs up catalog; local use remains available.	Continue locally.

7.11 Content and terminology rules

Use	Do not use
“Modelled position”	“Detected through your camera”
“Objects in this area”	“Exact object under the reticle”
“May be visible” / “Visibility unknown”	“Visible now” without sufficient evidence
“Elements updated 4 hours ago”	“Live” unless the data actually is live
“Catalogued debris”	“All debris”
“Use Manual Sky”	“Your device is unsupported” as a dead end
“Save object” / “Watch passes”	Clever labels that obscure the action
“Source: CeresTrak GP”	Unattributed educational facts

7.12 Low-fidelity screen direction

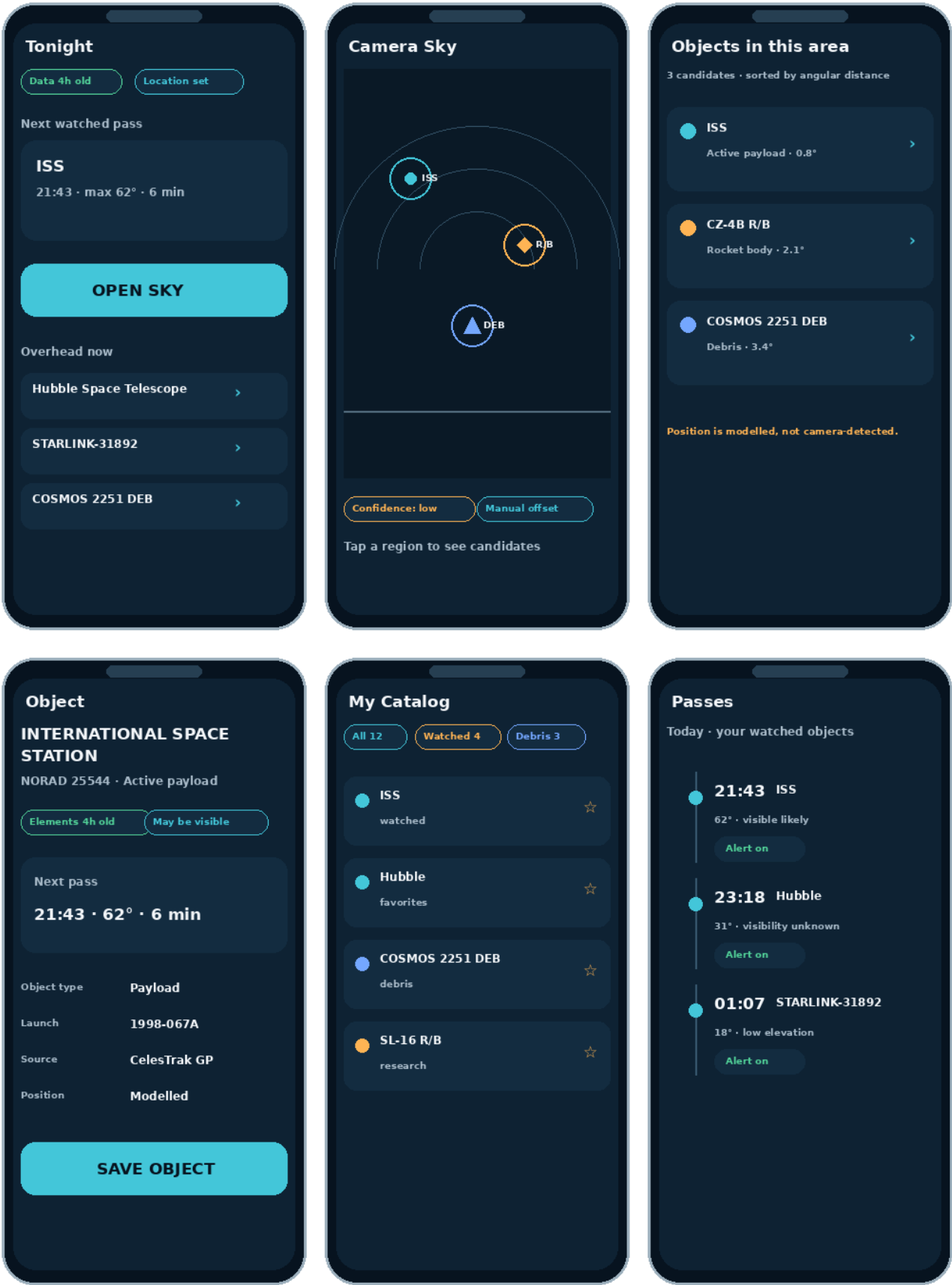


Figure 6 — Schematic screen direction for Tonight, Camera Sky, candidate selection, object detail, My Catalog and Passes. These are interaction contracts, not final visual comps.

8. Screen-by-screen specification

Detailed contracts for Claude Design and implementation. Every screen includes states, accessibility and acceptance criteria.

S-01 — Launch / bootstrap

Aspect	Screen contract
Purpose	Start safely and recover the last-known-good state.
Information hierarchy	Logo/wordmark, concise progress, data/package state only if delayed.
Primary/secondary actions	No permission request. If cache exists, open Home immediately and refresh in background.
Required states	Offline first install; corrupt cache; maintenance; mandatory migration.
Accessibility	Screen reader announces progress only when delayed; no animated starfield required.
Acceptance	Opens a usable state or explicit recovery; never loops indefinitely.

S-02 — Onboarding / value preview

Aspect	Screen contract
Purpose	Explain value in at most three focused steps.
Information hierarchy	Understand overhead → save objects → get pass alerts.
Primary/secondary actions	“Explore now” is primary; “How positions work” secondary.
Required states	Returning user skips; no account gate.
Accessibility	Headings, concise copy, skip, reduced motion.
Acceptance	User reaches Home without permissions or registration.

S-03 — Tonight / Home

Aspect	Screen contract
Purpose	Answer “what matters now?” and drive the core loop.
Information hierarchy	Global data status, next watched pass, Open Sky, overhead list, recent catalog.
Primary/secondary actions	Primary: Open Sky. Secondary: object/pass rows.
Required states	No location; no watched objects; offline; stale data; alert denied.
Accessibility	Logical heading order; rows are full-size buttons; time labels are spoken unambiguously.
Acceptance	New user understands the primary action; returning user sees next watched value.

S-04 — Sky mode chooser / permission primer

Aspect	Screen contract
Purpose	Let the user choose Camera or Manual Sky with informed consent.
Information hierarchy	Two mode cards, benefit, required resources, privacy/fallback.
Primary/secondary actions	Primary follows selected mode; “Not now” preserves Manual.
Required states	Permission previously denied; unsupported sensor; no camera.
Accessibility	No coercive copy; mode choice keyboard/screen-reader operable.
Acceptance	Denial never blocks Manual Sky.

S-05 — Camera Sky

Aspect	Screen contract
Purpose	Show modelled objects over camera while communicating uncertainty.
Information hierarchy	Compass/time, data status, reticle, markers, confidence, filters, Manual switch.
Primary/secondary actions	Primary interaction: tap region; secondary calibration/filter.
Required states	Calibrating; low confidence; no location; stale data; thermal/performance fallback.
Accessibility	Accessible alternative list is reachable; marker count/status announced on demand, not every frame.
Acceptance	No false single hit; stable enough for supported matrix gate.

S-06 — Manual Sky

Aspect	Screen contract
Purpose	Provide complete sky exploration without camera/motion.
Information hierarchy	Pan/tilt sky, time, horizon, markers, search, filters, candidate selection.
Primary/secondary actions	Drag/pan plus non-drag controls and reset.
Required states	No location; offline; high density.
Accessibility	Keyboard/assistive controls for movement; zoom buttons; list equivalent.
Acceptance	Can complete identify → detail → save flow.

S-07 — Candidate list

Aspect	Screen contract
Purpose	Resolve ambiguity after tap.
Information hierarchy	Candidate count, ordered rows, type, angular separation/confidence, model note.
Primary/secondary actions	Select candidate; close/back.
Required states	Zero candidates; many candidates; stale position.
Accessibility	Focus moves into sheet; rows are semantic buttons; close returns focus.
Acceptance	User can explain why multiple candidates appear.

S-08 — Object detail

Aspect	Screen contract
Purpose	Build trust and support save/watch.
Information hierarchy	Name/ID, type/status, source age, visibility, next pass, position wording, education/source details.
Primary/secondary actions	Primary: Save or Watch depending state.
Required states	Unknown name/type/visibility; decayed object; stale elements; offline.
Accessibility	Expandable sections use proper disclosure semantics; factual links have descriptive names.
Acceptance	No field implies unsourced certainty.

S-09 — Save / annotate

Aspect	Screen contract
Purpose	Capture the wedge with minimal friction.
Information hierarchy	Save state, optional tags, optional note, Watch toggle with notification

Aspect	Screen contract
	explanation.
Primary/secondary actions	Primary: Save.
Required states	Local-only; sync conflict later; note length; notification denied.
Accessibility	Labels not placeholders; note count; errors associated with fields.
Acceptance	Save succeeds locally without account/network.

S-10 — My Catalog

Aspect	Screen contract
Purpose	Make personal curation fast and rewarding.
Information hierarchy	Search, filter chips, watched/tag sections, sortable object rows.
Primary/secondary actions	Open object; toggle watch through explicit control; multi-select deferred.
Required states	Empty; offline; sync pending/conflict.
Accessibility	Chip state spoken; list supports large text; no swipe-only actions.
Acceptance	Find watched object and filter by tag within task threshold.

S-11 — Passes timeline

Aspect	Screen contract
Purpose	Show actionable upcoming passes for watched objects.
Information hierarchy	Date sections, pass time, maximum elevation, duration, visibility state, alert status.
Primary/secondary actions	Open pass; manage alert.
Required states	No passes; stale data; location missing; OS alert restrictions.
Accessibility	Local time zone explicit; screen reader wording includes date/time/angle.
Acceptance	Pass values use same element/version as detail or explain difference.

S-12 — Pass detail / alert destination

Aspect	Screen contract
Purpose	Prepare the user for one pass.
Information hierarchy	Countdown, rise/max/set, direction, confidence/freshness, object link, alert controls.
Primary/secondary actions	Open Sky near event; change alert.
Required states	Event passed; data refreshed; alert fired late.
Accessibility	No flashing countdown; reduced motion; time semantics.
Acceptance	Deep link arrives at correct pass/object.

S-13 — Search and filters

Aspect	Screen contract
Purpose	Control clutter and support expert lookup.
Information hierarchy	Name/NORAD search, type/status, visible-likely, watched, tags.
Primary/secondary actions	Apply/clear.
Required states	No result; partial/offline catalog.
Accessibility	Search label, clear button, filter summary; no hidden gesture.
Acceptance	Filters persist within session and are visible.

S-14 — Learn / education

Aspect	Screen contract
Purpose	Explain orbital environment with sourced facts.
Information hierarchy	Curated cards, tracked vs. modelled populations, object stories, source links and review date.
Primary/secondary actions	Open source / related object.
Required states	Content unavailable/offline; outdated review date.
Accessibility	Readable long-form, headings, alt text and no autoplay motion.
Acceptance	Every fact maps to claims registry/source ID.

S-15 — Data status

Aspect	Screen contract
Purpose	Make freshness and provider state diagnosable.
Information hierarchy	Package version, element age distribution, last refresh, source, retry, cache size.
Primary/secondary actions	Retry; view details.
Required states	Rate limited; source outage; validation failed; using last valid.
Accessibility	Status wording not colour-only; timestamps accessible.
Acceptance	Users and support can identify current data state.

S-16 — Settings / privacy / accessibility

Aspect	Screen contract
Purpose	Control permissions, local data, notifications, display and optional account.
Information hierarchy	Location mode, notification state, Manual default, reduced motion, text/system settings, export/delete, sync.
Primary/secondary actions	Contextual save or immediate switches.
Required states	OS setting changed; account unavailable; deletion pending.
Accessibility	Native controls/labels, confirmation for destructive action.
Acceptance	All data controls are discoverable and accurate.

S-17 — Optional sync / account

Aspect	Screen contract
Purpose	Enable backup without coercion.
Information hierarchy	Benefit, data categories, sign-in method, sync status, conflict resolution, delete account.
Primary/secondary actions	Enable/disable sync.
Required states	Offline, partial conflict, auth expiry.
Accessibility	No dark patterns; account deletion and local-only option.
Acceptance	Local catalog remains usable if sync fails or is disabled.

S-18 — Error / empty / offline patterns

Aspect	Screen contract
Purpose	Provide consistent recovery across the product.
Information hierarchy	What happened, what remains safe, one recovery action, fallback.
Primary/secondary actions	Retry or use cached/manual path.

Aspect	Screen contract
Required states	Provider failure, permission denial, parser error, unsupported device.
Accessibility	Error announced; focus placed at summary; no code-only message.
Acceptance	User data is preserved and next action is clear.

9. Accessibility and inclusive design

Accessibility is a release requirement and a core product fallback, not a post-design audit.

9.1 Conformance baseline

WCAG 2.2 is a W3C Recommendation organized under perceivable, operable, understandable and robust principles [E-04]. WCAG2ICT provides authoritative guidance for applying WCAG A/AA criteria to mobile and other non-web software [E-05]. OrbitMark therefore uses WCAG 2.2 AA as the measurable baseline while also testing platform-native behaviour with VoiceOver and TalkBack.

9.2 Product-specific accessibility requirements

ID	Requirement
A11Y-01	All core tasks have a non-camera and non-motion route through Manual Sky, lists and search.
A11Y-02	Sky canvas exposes an accessible summary and synchronized semantic object list; individual moving markers are not the only controls.
A11Y-03	Meaning uses text/shape/icon in addition to colour; object class and status are included in accessible names.
A11Y-04	All actionable targets meet the project minimum and have visible focus/pressed/disabled states.
A11Y-05	Dynamic type/text zoom to at least 200% does not clip critical content or hide actions; layouts reflow.
A11Y-06	Dragging has button/keyboard alternatives; sheets, dialogs, menus and disclosures follow platform/APG interaction patterns [E-06].
A11Y-07	Live updates are rate-limited and user-triggered for assistive technology; frame-by-frame marker motion is not announced.
A11Y-08	Reduced motion removes non-essential animation and replaces animated calibration with static guidance.
A11Y-09	Pass times and angles have meaningful spoken labels; abbreviations and symbols are expanded.
A11Y-10	Permission primers and errors use plain language and preserve focus after OS dialogs.
A11Y-11	Charts/educational visuals provide text equivalents and source links.
A11Y-12	Dark/night design still meets contrast. Optional red-light mode cannot become the only accessible theme.

9.3 Accessibility test matrix

Area	Tools / methods	Release evidence
Semantics	VoiceOver iOS, TalkBack Android, browser accessibility tree, native inspector.	Recorded task runs + defect log.
Keyboard/switch	External keyboard and switch-like linear navigation for PWA/native surfaces.	Focus order screenshots/video.
Text scaling	iOS Larger Text, Android font/display size, browser 200% zoom.	Screen matrix with no critical clipping.
Contrast/colour	Automated contrast check plus manual state review.	Token report and exception list.
Motion	Reduced-motion OS/browser setting.	Before/after behaviour list.
Touch/drag	Target-size audit; all drag actions tested with alternatives.	Component checklist.
Cognitive/plain language	Comprehension test for modelled position, freshness and confidence.	Participant evidence.

9.4 Accessibility stop-ship defects

- Core action unavailable to a screen-reader user or only available through Camera Sky.

- Focus trapped, lost behind a sheet, or returned to an unrelated element.
- Critical state conveyed only by colour or motion.
- Large text hides Save, Watch, Retry, permission fallback or data status.
- Canvas content has no semantic/list equivalent.
- Automated animation cannot be reduced or causes unusable jitter/flashing.

10. Security, privacy, policy and legal controls

The design minimizes collected data and prevents the education channel from silently becoming a child-data product.

10.1 Security baseline

OWASP MASVS is the approved mobile-security control framework [E-11]. The backend also uses applicable OWASP ASVS controls. Security scope is proportional to the product, but the absence of payment or sensitive health data does not remove obligations around authentication, location, notes, tokens, dependency integrity and deletion.

Control area	Mandatory design
Data storage	Store catalog and settings in platform-appropriate app storage; protect auth tokens in native secure storage; notes are never placed in logs or analytics.
Transport	TLS only; certificate/platform defaults; no secrets or provider credentials embedded in the client.
Authentication	Optional sync uses a proven identity flow. Sessions are revocable; account deletion invalidates tokens and deletes server records.
Authorization	Every user-scoped API query enforces ownership server-side; object metadata remains public/read-only.
Input/output	Validate tags/notes, lengths and encodings; escape rendered content; no executable markup from users.
Dependencies	Lockfiles, automated vulnerability scanning, SBOM and reviewed updates for Capacitor, React, satellite.js and server packages.
Build/release	Signed store builds, protected CI secrets, production flags, no DEV_MODE bypass in release artifacts.
Operations	Rate limits, audit events for sync/deletion, alerting on ingest failure, backup/restore tests.

10.2 Privacy data inventory

Data	MVP location	Server by default?	Retention/control
Precise current location	Device memory/local settings	No	User can use city/manual location, clear it, or deny permission.
Saved objects/tags/notes	Local catalog	No; optional sync only	Export/delete locally; synced data export/delete if enabled.
Watch/pass schedule	Local database and OS notification queue	No	User can disable watch/notifications and clear schedules.
Camera image	Live device view	No capture/upload by default	No photo persistence unless a separate user action is introduced and reviewed.
Motion/orientation	Device session	No	Ephemeral; no raw sensor telemetry.
Analytics identifier	Pseudonymous first-party ID	Only minimal events if enabled	No ad ID; rotation/deletion policy documented.
Account email/identity	Not required for local mode	Yes only for optional sync	Deletion and privacy policy required.

10.3 Children and education boundary

Apple and Google apply additional requirements when an app targets children or includes child-directed/social functionality [E-12][E-13]. The approved MVP therefore does not create child accounts, classroom rosters, free-form public communication or personalized advertising. Educators can facilitate use without collecting student identity. Any proposal to target children directly is an O item requiring legal review, revised privacy design, store-policy analysis and a new CDR.

10.4 Platform policy requirements

Policy area	Release action
Apple App Review	Review current Safety, Performance, Business, Design and Legal sections before submission [E-12].
Apple privacy labels	Inventory all first- and third-party data collection and keep declarations current.
Google Data safety	Complete declarations from the actual SDK/data inventory; privacy policy must match behaviour.
Permissions	Purpose strings and in-app primers must be specific, accurate and non-coercive.
Notifications	Ask only after the user creates a watch intent; document OS timing limitations.
Children/families	Declare target audience honestly; do not include children without the separate compliance program.
Subscriptions/monetization	Store billing and paywall scope are controlled by the business plan and require a separate implementation review.

10.5 Provider terms and content rights

Pre-launch legal gate

Official technical documentation confirms data formats and access patterns, but this CDR does not substitute for legal review of redistribution, caching, attribution and commercial-use terms for Celestrak, Space-Track, imagery, fonts or education content. The production ingest and public package are blocked until that review is recorded.

11. Quality attributes, operations and observability

Operational trust is part of the user experience because every position depends on time, source data and device state.

11.1 Quality attribute scenarios

Attribute	Stimulus	Required response
Reliability	Provider returns invalid/partial data.	Ingest quarantines the run; current package remains last valid; operations alert fires; UI shows last successful age.
Performance	User opens Sky with a large catalog.	Worker loads bounded set; main thread stays interactive; density control prevents marker flood.
Offline	User opens app in airplane mode after prior use.	Cached sky/catalog/passess work; data age is visible; refresh action explains offline state.
Privacy	User enables local passes but not sync.	No precise location or catalog leaves device; analytics contain no coordinates or note content.
Accessibility	User uses TalkBack and cannot hold device steadily.	Manual Sky/list path supports identify, detail, save and pass tasks.
Recoverability	Database migration or deploy fails.	Rollback restores previous app/API; last published package remains available; migration is not auto-destructive.
Maintainability	Provider adds large catalog IDs/new fields.	OMM/raw preservation and schema validation accept compatible fields; UI does not assume fixed width.

11.2 Observability model

Layer	Required signals
Ingest	Start/end, provider status, conditional request result, bytes, record count, invalid count, newest/oldest epoch, checksum, published package.
API	Request volume, latency, errors, package/cache hit, client version, rate-limit events; no precise coordinates in logs.
Client data	Package version, load/parse result, cache migration, age bucket, refresh error; no raw notes/location.
Sky quality	Mode, device family, confidence state transitions, render performance buckets, manual fallback usage; raw orientation is not retained.
Core loop	Open Sky, candidate shown, object detail, save, watch, pass viewed, alert scheduled/opened.
Privacy/security	Consent state, export/delete completion, auth failure, token revocation, dependency/build evidence.

11.3 Service-level objectives for beta

Numerical SLOs are set after M0/M1 instrumentation rather than invented before measurement. The release gate must nevertheless define and report: package freshness availability, API success/latency, crash-free sessions, failed notification scheduling, worker parse/propagation errors, and percentage of Sky sessions entering low-confidence/manual fallback.

11.4 Data/update failure UX

Condition	User-facing response	Operational response
No network, valid cache	"Offline · using data updated ..." Core functions continue.	No error alert; refresh on network return.
Provider unavailable	Last valid package remains; banner states age and source issue.	Alert after defined duration; exponential backoff.
Validation failure	Do not publish new package; user sees no false "updated" state.	Quarantine payload; error and sample record logged.

Condition	User-facing response	Operational response
Stale beyond alert threshold	Pass alert generation is blocked or explicitly labelled per approved rule.	Escalate source/ingest issue.
Partial object metadata	Show known fields and "Unknown"; no fabricated fallback.	Track completeness metric.

12. Verification and validation plan

The evidence package required to move from design approval to beta and production.

12.1 Verification layers

Layer	What is verified	Primary evidence
Static/contract	Schemas, IDs, source attribution, API compatibility, claims strings, accessibility lint.	CI reports and contract tests.
Unit	Coordinate transforms, ranking, age states, filters, catalog operations, notification scheduling logic.	Automated tests.
Reference numerical	SGP4/topocentric output for fixed OMM/time/location against independent implementation.	Golden vector report.
Integration	Provider → ingest → DB → package → client cache → worker → UI; failure/rollback paths.	Environment test logs.
Device field	Orientation, permission, camera, performance, thermal, rotation and manual offset across device matrix.	Structured field protocol/video/data.
Usability	Newcomer, enthusiast and educator tasks; comprehension of modelled/uncertainty/freshness.	Moderated session findings.
Accessibility	VoiceOver/TalkBack, dynamic text, keyboard, contrast, reduced motion and non-AR parity.	Audit report.
Security/privacy	MASVS/ASVS checklist, threat model, dependency scan, data-flow and deletion test.	Security evidence pack.
Release/claims	Store metadata, screenshots, privacy declarations, sources and prohibited-claim scan.	Signed release checklist.

12.2 Device test matrix

Dimension	Minimum beta coverage
iOS	At least three supported iPhone generations, including the oldest supported model and one current model.
Android	At least four devices across two major vendors, including a device with weak/noisy compass behaviour and the oldest supported Android baseline.
Orientation	Portrait, safe landscape/manual mode where supported, screen rotation and interruption/resume.
Environment	Open outdoor sky, urban/magnetic interference, indoors/manual fallback, low light and daylight.
Permissions	First request, deny, deny permanently, OS settings change, revoke after scheduling.
Network	Online, slow, offline, source outage, corrupted/partial package.
Accessibility	VoiceOver/TalkBack, largest text, reduced motion, high contrast/platform settings, external keyboard.

12.3 Usability test protocol

Task	Success evidence	Critical misconception
Open a usable sky without granting camera	User chooses Manual Sky and sees modelled objects.	Belief that the app is unusable without permissions.
Find what is in a tapped region	User opens candidate list and selects deliberately.	Belief that reticle equals exact identification.
Understand object status	User states type, source age and modelled nature.	Belief that invisible/untracked debris is displayed individually.
Save and watch	User saves, optionally notes/tags, enables	Belief that registration is mandatory.

Task	Success evidence	Critical misconception
	watch and understands notification request.	
Find next pass	User locates next watched pass and can explain visibility state.	Belief that alert timing/visibility is guaranteed.
Recover from low confidence/offline	User switches to Manual or uses cached data.	Dead end or loss of trust.

12.4 Design review scorecard

Category	Weight	Pass condition
Task clarity and hierarchy	20	One primary action; no ambiguous navigation or jargon.
Truth, provenance and claims	15	Model/freshness/confidence visible; no prohibited wording.
Accessibility	20	No critical defect; equivalent non-AR route; tested states.
State completeness	10	Loading, empty, offline, denied, stale, error, success and destructive states specified.
Visual system coherence	10	Tokens/components consistent and subject-specific; no generic template drift.
Responsive/platform fit	10	Safe areas, dynamic text, device sizes and native conventions handled.
Performance feasibility	10	Design respects object-density, animation and rendering budgets.
Privacy/security	5	No unnecessary collection or hidden permission/account pressure.

A screen set requires $\geq 85/100$ and no zero-tolerance failure in truth/claims, accessibility, privacy or core task completion. The score is a project governance mechanism, not an industry benchmark.

13. Delivery roadmap and release gates

Work is sequenced behind evidence, not calendar optimism.



Figure 7 — Approved milestone and evidence-gate flow.

Milestone	Scope	Deliverables	Exit gate
M0	Evidence spike	Provider OMM ingest; large-ID schema; satellite.js worker; fixed-vector validation; camera/motion/permission spike; initial performance profile.	G0: reference vectors pass; OMM package valid; two iOS + two Android devices demonstrate permission and orientation path; no unresolved architecture blocker.
M1	Core sky	Home, Manual Sky, Camera Sky, orientation abstraction, confidence state, candidate list, object detail, freshness/offline.	G1: supported-device field test; Manual parity; accessibility smoke; no misleading claim; performance budget fixed.
M2	Curation wedge	Local catalog, save/note/tag/watch, pass engine, local notifications, Catalog and Passes.	G2/MVP: core task success; alert schedule/reschedule test; local-first privacy audit; analytics core-loop evidence.
M3	Trust and beta	Education/source registry, data status, optional sync, share card, production operations and external reviews.	Beta gate: legal/provider review; security and accessibility audit; claims/store/privacy evidence; rollback/incident runbook.
M4	Post-validation depth	3D globe, richer histories, facilitator tools and experiments.	Only if curation/retention evidence supports them. New data/privacy scope reviewed.
M5	Research track	Plate solving or native advanced tracking.	Separate feasibility, privacy, battery and accuracy CDR.
M0.5	Orbital data authority implementation	Ingest state machine, immutable schema, analyst exclusion, package manifest/atomic publish, client checksum activation, frame adapter, reference fixtures and operations dashboard.	G0.5: all Appendix G P0 contracts and failure injections pass before broad UI implementation.

13.1 Definition of Done for every feature

- Requirement IDs and screen contract are referenced in the pull request/design file.
- Default, loading, empty, offline, denied, stale, error and accessibility states are implemented where applicable.
- Automated tests and manual evidence cover the agreed verification method.
- No user-facing factual claim lacks a source or approved claims-registry entry.
- Analytics event and privacy impact are reviewed; no precise location or note content is introduced.
- Visual regression is reviewed at supported breakpoints/text scales.
- Documentation, migration, rollback and operational signals are updated.

13.2 Change control

Any change to the wedge, audience, data provider, location flow, Camera Sky accuracy claim, account requirement, social feature, child-directed use, pass-computation location or OMM canonical model requires an impact note and Council re-review. Changes that introduce operator-grade functionality require a new product thesis and CDR.

14. Risk register, open items and CDR actions

Release-blocking risks are converted into named actions with evidence owners.

Risk	Likelihood	Impact	Failure mode	Control
R-01	High	High	Camera Sky does not align consistently across devices.	Orientation abstraction, manual parity, confidence state, native plugin escape hatch and G1 field gate.
R-02	High	High	Product feels like another tracker; curation loop is secondary.	Home/Catalog/Passes hierarchy, activation instrumentation and M2 usability gate.
R-03	Med	High	Provider terms do not permit the intended redistribution/caching pattern.	Legal review before public package; alternative client/provider architecture retained.
R-04	Med	High	Users interpret modelled overlay as camera detection.	Persistent wording, candidate list, education, claims scan and comprehension testing.
R-05	Med	High	Large catalog or worker design causes battery/performance issues.	Bounded subset, controlled cadence, profiling, feature flags and density limits.
R-06	Med	High	Location or notes leak through logs, analytics or sync.	Local-first design, data inventory, no-location logging, privacy tests and MASVS review.
R-07	Med	Med	Local notifications are delayed/removed by OS settings.	Rolling schedule, state inspection, reschedule on activation and honest copy.
R-08	Med	Med	Stale elements generate misleading passes.	Visible age, last-valid protection, configurable alert-age policy and source monitoring.
R-09	Low/Med	High	Education scope enters children/families compliance unexpectedly.	No child accounts/social MVP; target-audience declaration and separate review.
R-10	Med	Med	AI-assisted design produces visually attractive but non-compliant screens.	CDR authority, skill order, screen checklist, accessibility and visual regression gates.
R-11	Med	High	Catalog-number reuse or identity drift attaches user annotations to a different physical object.	Separate object_key, analyst/temporary exclusion, lineage fixtures and identity gate.
R-12	Med	High	Frame or unit ambiguity produces plausible but wrong sky positions.	TEME/unit naming, single adapter, independent reference vectors and code-review checklist.
R-13	Med	High	Partial/corrupt package activation removes the last usable catalog.	Immutable package, checksum, atomic current pointer and client last-known-good rollback.

14.1 Open items

ID	Owner domain	Decision required	Due gate
O-001	Legal	CelesTrak/Space-Track redistribution, caching, attribution and commercial-use review.	Before M3 public beta
O-002	Architecture	Whether web/Capacitor motion exposes sufficient heading quality; native plugin decision.	G1
O-003	Product	Supported OS/device baseline and device exclusion policy.	End of M0

ID	Owner domain	Decision required	Due gate
O-004	Engineering	Empirical element-age policy for pass alerts by orbit/object class.	M1/M2
O-005	Security	Identity provider and sync conflict model.	Before optional sync
O-006	Accessibility	Independent mobile accessibility audit provider and scope.	Before beta
O-007	Content	Education editorial owner, review cadence and source registry process.	M3
O-008	Brand	Final product name/trademark and font licensing.	Before store assets
O-009	Operations	Data retention for element history and telemetry.	Before production
O-010	Commercial	Paywall placement and free/pro boundaries without damaging activation.	Separate monetization design review
O-011	Orbit engineering	Final element-age guardrails by orbit class and user-facing degradation wording, based on validation evidence rather than a universal age limit.	G0.5/G1
O-012	Performance	Working-set sizes and propagation cadences for each supported device tier.	End of M0.5
O-013	Product/Legal	Whether any analyst or SupGP objects may be exposed after identity, terms and source-labelling review.	Post-MVP/new ADR

14.2 Mandatory CDR actions

Action	Required work	Owner	Due
A-01	Update all architecture diagrams and prose to remove "satellite.js WASM," per-frame IndexedDB and server-default pass/location flow.	Architecture owner	Before coding
A-02	Create OMM-first schema and golden test vectors including identifiers >99,999.	Backend/orbit	M0
A-03	Build orientation spike and field protocol before visual polish.	Mobile	M0/G1
A-04	Create source/claims registry used by product copy, education content, store assets and share cards.	Product/content	M1
A-05	Create local-first data-flow diagram and privacy inventory; prohibit precise coordinates in telemetry.	Privacy/security	M1
A-06	Implement Manual Sky and semantic list before Camera Sky is considered complete.	UX/frontend	M1
A-07	Create design tokens/components and screen states in a controlled DESIGN.md/JSON source.	Design authority	M1
A-08	Set up visual, accessibility, contract and device verification in CI/release evidence.	QA	M1–M3
A-09	Complete legal/provider and store-policy review.	Founder/counsel	M3
A-10	Run external accessibility and security reviews before public beta.	Founder/QA	M3
A-11	Implement Appendix G ingest, identity, package and numerical contracts before scaling screen	Backend/orbit/QA	M0.5

Action	Required work	Owner	Due
	implementation.		
A-12	Create golden OMM, propagation, coordinate and pass fixtures with independent expected outputs and failure cases.	Orbit/QA	M0.5
A-13	Design Data Status, stale/clock-skew, identity-limited and package-fallback states from Appendix G.	Design authority/UX	M1

15. Claude Design and implementation protocol

How Claude Design / Claude Code must consume this CDR without inventing scope or weakening controls.

15.1 Skill stack — authority order

Order	Skill / method	Approved role	Limit
1	OrbitMark CDR authority	Controls requirements, ADRs, screen contracts, tokens, claims and gates.	May be changed only through recorded change control.
2	Anthropic official frontend-design	Generate a distinctive, subject-grounded visual direction and critique generic defaults [E-18].	Must comply with CDR tokens, content and accessibility.
3	Project skill: orbitmark-design-authority	Inject product principles, screen checklist, source wording and token contract.	Project-specific; must cite requirement IDs.
4	Project skill: orbitmark-a11y-review	Review WCAG2ICT/platform semantics, focus, text scale, colour, motion and non-AR parity.	Cannot self-certify final conformance.
5	Project skill: orbitmark-claims-gate	Scan copy and visuals for detection, accuracy, live-data, visibility and debris overclaims.	Blocks merge on prohibited language.
6	Web app / run / verify testing skills	Launch, interact, capture screenshots and verify real behaviour rather than code-only confidence [E-19].	Device-native behaviour still needs physical-device testing.
7	Community UI/UX skills (optional)	Generate references, palettes or pattern alternatives.	Advisory only; never authoritative or accepted without evidence.

15.2 Required design sequence

34. Read this CDR and output the requirement IDs, ADRs and screen contracts relevant to the requested screen or flow.
35. State only unresolved assumptions. Do not invent features, data fields, claims or permissions.
36. Create a compact design plan: visual thesis, token use, hierarchy, content, signature element, accessibility and edge states.
37. Produce at least two structurally different low-fidelity variants when the information architecture is not fixed; compare against task clarity and truth requirements.
38. Create the selected high-fidelity design using real OrbitMark copy and data, not lorem ipsum.
39. Add every required state: loading, empty, offline, denied, stale, error, success, destructive and accessibility/large-text where applicable.
40. Run the design scorecard, claims gate and accessibility review. Resolve defects before implementation.
41. Implement with semantic components and the approved token source. Do not reproduce the mock as inaccessible absolute-positioned pixels.
42. Run/verify the application, capture screenshots at required devices/text scales and attach evidence to the work item.
43. Update DESIGN.md, component documentation and requirement traceability before merge.

15.3 Claude Design master brief

You are the product design authority implementing OrbitMark under OM-CDR-001. Treat the CDR as controlling. OrbitMark is an honest, model-based orbital-object companion centred on saving, annotating, watching and revisiting personal objects. Camera Sky is optional and never described as camera detection. Manual Sky and semantic lists must complete every core task.

For the requested screen or flow:

1. List the controlling requirement IDs, ADRs and screen contract.
2. State unresolved assumptions; do not invent scope.
3. Use the approved night-instrument visual thesis and token system. The uncertainty ring is the signature honesty motif. Avoid generic neon sci-fi, purple gradients, game HUDs and excessive pills.
4. Show real content, source age, model wording, confidence and visibility unknowns.
5. Specify default, loading, empty, offline, denied, stale, error, success, large-text, reduced-motion and screen-reader behaviour.
6. Ensure one primary action, plain verb-first copy, visible focus, non-colour status, target size and non-drag alternatives.
7. Produce a rationale tied to user jobs and a scorecard. Do not call the design complete until the claims and accessibility gates pass.

15.4 Required design deliverables

Deliverable	Minimum content
DESIGN.md / token source	Brand thesis, palette, typography, spacing, radius, motion, object-marker semantics, component states and accessibility rules.
Screen file	All S-01...S-18 screens applicable to release, with state variants and requirement IDs.
Clickable prototype	Core loop: Home → Sky → candidates → detail → save/watch → Passes → alert deep link; Manual route included.
Component inventory	Semantic components, props/states, accessibility contract and responsive behaviour.
Copy/source registry	Every factual string, source ID, review date and prohibited alternatives.
Design QA report	Scorecard, contrast, large text, keyboard/screen reader, reduced motion, device screenshots and unresolved defects.

15.5 Implementation guardrails

- No direct styling values outside the approved token source except documented temporary experiments.
- No canvas-only interaction without a synchronized semantic list/control path.
- No visual marker or chip without an accessible name/state.
- No exact angle/accuracy display unless the value is actually measured and validated.
- No permission call outside the approved contextual flow.
- No analytics event containing coordinates, note text, search text or raw orientation.
- No AI-generated educational fact shipped without source registry approval.
- No screenshot/store asset created from a stale or non-compliant screen state.

16. Final decision and approval record

The design plan is authorized subject to the conditions below.

16.1 CDR decision

APPROVE WITH CONDITIONS

OrbitMark may proceed to M0 and M1 implementation under this CDR. The approved product is a local-first, annotation-centred, model-labelled orbital-object companion. The conditions are part of the approval, not optional recommendations.

16.2 Conditions of approval Version 1.1 additionally conditions production-capable orbital ingestion, sky position and pass prediction on Appendix G and Gate G0.5.

Condition	Release obligation
COA-01	OMM-first, large-ID schema and independent numerical validation completed at G0.
COA-02	Camera Sky passes the supported-device field gate or is limited/disabled by device with Manual Sky preserved.
COA-03	Manual Sky and semantic list achieve full P0 task parity and accessibility gate.
COA-04	Precise location remains local by default; any server location flow requires privacy change review.
COA-05	Provider terms, app-store privacy/policy, security and external accessibility review are signed before public beta.
COA-06	Marketing, UI, education and share assets pass the source/claims registry with no camera-detection or pinpoint claim.
COA-07	Ingestion is separated from web workers and demonstrated idempotent/observable.
COA-08	The product loop demonstrates that users can save/watch and return; otherwise scope is reconsidered before expansion.
COA-09	Appendix G is implemented and its G0.5 evidence pack is approved before Camera Sky or pass alerts are treated as production-capable.
COA-10	Analyst-range and temporary identities remain excluded from persistent user features until a lineage-safe identity ADR is approved.

16.3 Approval signatures

Role	Name	Decision	Date / signature
Founder / Product Owner	Rotem	☐ Approve ☐ Approve with changes ☐ Reject	
Engineering authority		☐ Approve ☐ Approve with changes ☐ Reject	
UX / accessibility authority		☐ Approve ☐ Approve with changes ☐ Reject	
Security / privacy reviewer		☐ Approve ☐ Approve with changes ☐ Reject	
Legal / data-terms counsel		☐ Approve ☐ Approve with changes ☐ Reject	

16.4 Final professional caveat

This CDR is an evidence-controlled product and engineering design document. It is suitable as an internal design authority and investor/partner diligence artifact, but it is not a legal opinion, accessibility certification, orbital-safety service, security attestation or guarantee of commercial performance. Those claims require the named external evidence and approvals.

Appendix A. Requirements traceability matrix

Mapping from requirements to architecture, screens and release evidence.

Req.	Pri.	Primary screens	ADRs	Verification	Gate
FR-001	P0	S-03/S-05/S-06	ADR-001/002/003/004/005/006/007/012	UX inspection + clean-install test	M3/Beta
FR-002	P0	S-03/S-05/S-06	ADR-001/002/003/004/005/006/007/012	Usability + offline test	M3/Beta
FR-003	P0	S-03/S-05/S-06	ADR-001/002/003/004/005/006/007/012	Functional parity test	G1
FR-004	P0	S-03/S-05/S-06	ADR-001/002/003/004/005/006/007/012	Permission-flow test	G1
FR-005	P0	S-05/S-07/S-08	ADR-001/002/003/004/005/006/007/012	Numerical validation	G0
FR-006	P0	S-05/S-07/S-08	ADR-001/002/003/004/005/006/007/012	Visual/accessibility test	G1
FR-007	P0	S-05/S-07/S-08	ADR-001/002/003/004/005/006/007/012	Interaction test	G1
FR-008	P0	S-05/S-07/S-08	ADR-001/002/003/004/005/006/007/012	Content contract test	G1
FR-009	P0	S-09/S-10/S-11/S-12	ADR-001/002/003/004/005/006/007/012	Persistence test	G2
FR-010	P0	S-09/S-10/S-11/S-12	ADR-001/002/003/004/005/006/007/012	Cross-engine test	G2
FR-011	P0	S-09/S-10/S-11/S-12	ADR-001/002/003/004/005/006/007/012	Platform test	G2
FR-012	P0	S-15/S-18	ADR-001/002/003/004/005/006/007/012	Failure injection test	G1
FR-013	P0	S-15/S-18	ADR-001/002/003/004/005/006/007/012	Airplane-mode test	G1
FR-014	P0	S-05/S-07/S-08	ADR-001/002/003/004/005/006/007/012	State-machine test	G1
FR-015	P0	Roadmap	ADR-001/002/003/004/005/006/007/012	Accessibility audit	G1
FR-016	P1	S-09/S-10/S-11/S-12	ADR-001/002/003/004/005/006/007/012	Functional test	G2
FR-017	P1	S-14	ADR-001/002/003/004/005/006/007/012	Editorial/source review	M3/Beta
FR-018	P1	S-16/S-17	ADR-001/002/003/004/005/006/007/012	Privacy test	M3/Beta
FR-019	P1	S-16/S-17	ADR-001/002/003/004/005/006/007/012	Sync conflict test	M3/Beta
FR-020	P1	S-08/share	ADR-001/002/003/004/005/006/007/012	Claims/privacy review	M3/Beta
FR-021	P1	S-15/S-18	ADR-001/002/003/004/005/006/007/012	Operations test	M3/Beta
FR-022	P2	Roadmap	ADR-001/002/003/004/005/006/007/012	Roadmap gate	M3/Beta
FR-023	P2	Roadmap	ADR-001/002/003/004/005/006/007/012	New CDR	M3/Beta
NFR-001	P0	Cross-cutting	ADR-003/004/005/006/009/011/012	Automated reference-vector suite	G0
NFR-002	P0	Cross-cutting	ADR-003/004/005/006/009/011/012	Code/content review	G1
NFR-003	P0	Cross-cutting	ADR-003/004/005/006/009/011/012	Performance profile	G1
NFR-004	P0	Cross-cutting	ADR-003/004/005/006/009/011/012	Architecture/code review	G0
NFR-005	P0	Cross-cutting	ADR-003/004/005/006	Schema/property tests	G0

Req.	Pri.	Primary screens	ADRs	Verification	Gate
			/009/011/012		
NFR-006	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Integration/failure test	G0
NFR-007	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Privacy data-flow audit	Beta
NFR-008	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Independent audit + automated checks	G1
NFR-009	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Security checklist + penetration test	Beta
NFR-010	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Claims registry audit	Beta
NFR-011	P0	Cross-cutting	ADR-003/004/005/006 /009/011/012	Data inventory review	Beta
NFR-012	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Pseudo-localization test	G1
NFR-013	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Platform accessibility test	G1
NFR-014	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Failure injection test	G1
NFR-015	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Contract test	Beta
NFR-016	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Deployment inspection	Beta
NFR-017	P1	Cross-cutting	ADR-003/004/005/006 /009/011/012	Release evidence pack	Beta
FR-024	P0	S-15/S-18	ADR-013–020	Provider integration test	G0.5
FR-025	P0	S-15/S-18	ADR-013–020	Schema/range/failure injection suite	G0.5
FR-026	P0	S-15/S-18	ADR-013–020	Database immutability test	G0.5
FR-027	P0	Cross-cutting	ADR-013–020	Identity fixture test	G0.5
FR-028	P0	S-15/S-18	ADR-013–020	Package atomicity test	G0.5
FR-029	P0	S-15/S-18	ADR-013–020	Client rollback test	G0.5
FR-030	P0	S-05/S-06/S-07/S-08	ADR-013–020	Numerical pipeline test	G0.5
FR-031	P0	S-05/S-06/S-07/S-08	ADR-013–020	Worker scheduling/performance test	G0.5
FR-032	P0	S-09/S-11/S-12	ADR-013–020	Pass-event cross- engine test	G2
FR-033	P0	S-05/S-06/S-07/S-08	ADR-013–020	Domain/content contract test	G1
FR-034	P0	S-15/S-18	ADR-013–020	Failure-state mapping test	G0.5
FR-035	P1	S-15/S-18	ADR-013–020	Supportability test	G1
NFR-018	P0	Cross-cutting	ADR-013–020	Time parsing property tests	G0.5
NFR-019	P0	Cross-cutting	ADR-013–020	Static contract review	G0.5
NFR-020	P0	Cross-cutting	ADR-013–020	Secret scanning + architecture test	G0.5
NFR-021	P0	Cross-cutting	ADR-013–020	Transaction/failure injection test	G0.5
NFR-022	P0	Cross-cutting	ADR-013–020	Golden fixture suite	G0.5
NFR-023	P0	Cross-cutting	ADR-013–020	Clock-skew simulation	G1
NFR-024	P1	Cross-cutting	ADR-013–020	Provider policy test	G0.5
NFR-025	P1	Cross-cutting	ADR-013–020	Profile + code inspection	G1
NFR-026	P1	Cross-cutting	ADR-013–020	Fixture coverage report	G0.5

Appendix B. API and data contract

Implementation-level baseline; detailed OpenAPI/schema files must be generated from this contract.

B.1 Public API surface

Method/path	Purpose	Key contract
GET /api/v1/element-packages/current	Return current package metadata.	Version, checksum, source, published_at, min/max epoch, record_count, URL/delta and compatibility.
GET /api/v1/element-packages/{version}	Download compressed immutable package.	OMM-compatible records; ETag/checksum; cache control.
GET /api/v1/objects/{catalog_id}	Return object metadata and source fields.	ID as string-safe value; unknown values are null, never fabricated.
GET /api/v1/status/data	Expose public data status.	Last successful ingest, source, package age and degradation state; no secrets.
POST /api/v1/sync/catalog	Optional authenticated sync batch.	Version/tombstones/idempotency; notes/tags validated; no location.
GET /api/v1/sync/catalog	Optional authenticated restore/delta.	Cursor/version; deletion markers.
DELETE /api/v1/me	Delete optional account and synced catalog.	Re-authentication and completion record.
GET /api/v1/element-packages/{version}/manifest	Return immutable package manifest without package body.	Format/client version, source set, generated_at, epoch distribution, record counts, checksum, compression and optional delta base.
GET /api/v1/sources	Expose non-secret source and attribution metadata.	Source ID, provider, dataset class, last success, terms/attribution label and public status.
GET /api/v1/status/ingest	Authenticated operations-only ingest status.	Run state, HTTP class, quarantine counts, package candidate, backoff and error codes; no provider credentials.

B.2 Core data entities

Entity	Required fields
objects	catalog_id BIGINT; object_name; international_designator; object_type; operational_status; launch_date; actual_decay_date; country/operator where sourced; metadata_source; updated_at.
element_sets	id; catalog_id; epoch; source; source_record_id; raw_omm JSONB; normalized SGP4 fields; fetched_at; ingest_run_id; checksum; validation_status.
ingest_runs	id; source; started_at; finished_at; status; response metadata; record_count; invalid_count; package_version; error summary.
sync_user_objects	user_id; catalog_id; tags; note; watch; client_version; record_version; updated_at; deleted_at. No precise location.
client package metadata	version; format_version; source list; generated_at; checksum; min/max epoch; record_count; required_client_version.
object_identities	object_key UUID; source_namespace; catalog_id; identity_class; valid_from; valid_to; lineage_status; public_eligible; reason.
quarantine_records	id; ingest_run_id; source_record_key; raw payload reference; validation codes; first_seen; last_seen; disposition.
package_manifests	version; format_version; required_client_version; source_set; generated_at; record_count; min/max epoch; checksum; size; status; published_at.

B.3 Client event/message contract

Message	Payload principle
worker:init	Package version and approved observer settings; no UI object references.

Message	Payload principle
worker:select	Time, horizon, filters, watched IDs and density budget.
worker:positions	Bounded object list with catalog ID, altitude, azimuth, optional visibility flags and source age metadata.
worker:candidates	Angular center/radius and ordered candidates with angular separation.
worker:passes	Watched IDs, date window and local observer input; returns rise/max/set and validation metadata.
orientation:state	Normalized heading/elevation/roll where available, quality state, manual offset and timestamp; raw data not logged.
worker:packageActivate	Verified package version, manifest and normalized records; worker returns accepted/rejected with reason code.
worker:diagnostics	Non-sensitive counters: package, parsed count, active bands, last compute duration, failure-code buckets and clock state.
worker:passRequest	Object key, element-set identity, UTC window, observer and horizon mask; returns rise/max/set plus solver and freshness metadata.

Appendix C. Analytics event dictionary

Minimal measurement of the wedge and quality; no precise location, notes or raw sensor data.

Event	Allowed properties
app_open	source, app_version, returning_flag
home_open_sky	selected_mode
permission_primer_view	permission, context
permission_result	permission, result
sky_session_start	mode, cache_age_bucket, device_family
confidence_state_change	from, to, reason_bucket
candidate_list_open	candidate_count_bucket, mode
object_detail_open	object_type, entry_surface
object_save	object_type, local_or_sync
object_watch_enable	object_type, notification_state
pass_list_view	watched_count_bucket, data_age_bucket
notification_schedule_result	result, platform, reason_bucket
notification_open	lead_time_bucket
manual_fallback	origin, reason_bucket
data_refresh_result	result, source, age_bucket
offline_mode_used	surface
a11y_mode_signal	reduced_motion/text_scale_bucket if locally detectable and policy-approved
export_complete	local_or_sync
delete_complete	local_or_account

Forbidden analytics payload

Exact latitude/longitude, free-form notes/tags/search strings, email, camera frames, raw heading/orientation, full catalog contents, notification body text and provider credentials.

Appendix D. Design token reference

Machine-readable values should be maintained in the repository and generated into platform styles.

D.1 Token JSON example

```
{
  "color": {
    "bg": {
      "canvas": "#07131F",
      "surface": "#0E2233",
      "panel": "#132C40"
    },
    "text": {
      "primary": "#F4F7FA",
      "secondary": "#A9BBC9"
    },
    "action": {
      "primary": "#43C6D9",
      "focus": "#74A7FF"
    },
    "status": {
      "warning": "#FFB454",
      "danger": "#FF6B6B",
      "success": "#53D49A"
    }
  },
  "space": {
    "1": 4,
    "2": 8,
    "3": 12,
    "4": 16,
    "6": 24,
    "8": 32
  },
  "radius": {
    "control": 12,
    "panel": 16,
    "sheet": 20
  },
  "motion": {
    "fast": 150,
    "standard": 220
  },
  "target": {
    "minimum": 44
  }
}
```

D.2 Contrast verification

Foreground	Background	Approx. contrast	Use
#F4F7FA	#07131F	17.4:1	Primary text
#A9BBC9	#0E2233	8.2:1	Secondary text
#43C6D9	#132C40	7.1:1	Cyan text/outline
#74A7FF	#132C40	6.0:1	Focus/secondary action
#FFB454	#132C40	8.1:1	Warning text
#FF6B6B	#132C40	5.2:1	Error text

Foreground	Background	Approx. contrast	Use
#53D49A	#132C40	7.7:1	Success text

Values are calculated from the proposed sRGB tokens. Final components must also verify contrast for filled states, opacity, disabled controls, images/camera backgrounds and platform rendering.

Appendix E. Source register

Primary and official sources used to control factual claims in this CDR. Accessed 19 June 2026 unless noted.

ID	Class	Source	Reference
I-01	Internal	OrbitMark Business Plan v1.0, June 2026	Commercial and product thesis; supplied file.
I-02	Internal	Orbital Debris AR App PDR v1	Preliminary technical/product review; supplied HTML.
I-03	Internal	11_DIAGRAMS.md	Initial architecture and interaction diagrams; supplied Markdown.
E-01	Primary technical	CelesTrak — GP data formats / OMM migration	Open official source
E-02	Primary technical	Space-Track — documentation for GP/GP_History and OMM formats	Open official source
E-03	Primary implementation	satellite.js repository and documentation	Open official source
E-04	Normative standard	W3C WAI — WCAG 2 overview / WCAG 2.2	Open official source
E-05	Authoritative guidance	W3C WAI — WCAG2ICT	Open official source
E-06	Authoritative guidance	W3C WAI — ARIA Authoring Practices Guide	Open official source
E-07	Platform guidance	Apple Human Interface Guidelines — Augmented reality	Open official source
E-08	Platform guidance	Apple Human Interface Guidelines — Accessibility	Open official source
E-09	Platform guidance	Material Design 3 — Accessible design	Open official source
E-10	International standard	ISO 9241-210:2019 — Human-centred design for interactive systems	Open official source
E-11	Security standard	OWASP Mobile Application Security Verification Standard	Open official source
E-12	Platform policy	Apple App Review Guidelines and privacy details	Open official source
E-13	Platform policy	Google Play Families / Data safety policies	Open official source
E-14	Primary implementation	Capacitor Motion API	Open official source
E-15	Web specification	W3C Device Orientation and Motion	Open official source
E-16	Primary implementation	Capacitor Local Notifications API	Open official source
E-17	Primary implementation	APScheduler documentation	Open official source
E-18	Official AI skill	Anthropic frontend-design skill	Open official source
E-19	Official AI tooling	Anthropic Claude Code skills documentation	Open official source
E-20	Primary technical	Vallado, Crawford, Hujsak & Kelso — Revisiting Spacetrack Report #3, AIAA 2006-6753	https://www.celestrak.org/publications/aiaa/2006-6753/AIAA-2006-6753.pdf
E-21	Primary technical	CelesTrak — Orbital Coordinate Systems, Part II (look-angle calculation)	https://www.celestrak.org/columns/v02n02/
E-22	Normative standard	CCSDS 502.0-B-3 — Orbit Data Messages, including OMM	https://ccsds.org/Pubs/502x0b3e1.pdf

Appendix F. Glossary

Controlled terminology used by product, design and engineering.

Term	Controlled meaning
Catalogued object	A public catalog record with an identifier and element data; not all physical debris is catalogued.
OMM	Orbit Mean-Elements Message, used here as the canonical structured orbital-element format.
SGP4	The propagation model associated with GP/TLE/OMM element sets.
Modelled position	A computed expected direction from element data, time and observer location; not a camera detection.
Camera Sky	Camera background plus orientation-based model overlay.
Manual Sky	Synthetic sky controlled by user input, independent of camera/motion permission.
Candidate list	Objects falling within the selected angular/uncertainty region.
Confidence state	Qualitative product state describing orientation usability; not an invented accuracy guarantee.
Element age	Time since the orbital element epoch, distinct from client fetch time.
Data freshness	Combined view of source/ingest/package state and element age.
Watched object	A saved object for which the user requests pass monitoring/alerts.
Claims registry	Controlled list connecting each factual product/marketing statement to source and review date.
Release evidence pack	Tests, audits, screenshots, policy declarations and approvals required at a gate.
GP element set	General Perturbations mean-element data intended for SGP4/SDP4 propagation; it is not a live position measurement.
TEME	True Equator Mean Equinox, the frame convention associated with the SGP4 state used by this CDR.
ECF	Earth-centred, Earth-fixed coordinates used after applying Earth rotation for observer-relative calculation.
Element epoch	The UTC instant at which the mean elements apply; distinct from fetch time and package publication time.
Object key	OrbitMark's stable internal identity, separate from a provider catalog number.
Analyst object	A Space-Track object in a reusable analytical range with insufficient stable catalog identity; excluded from persistent MVP user features.
Last-known-good package	The most recent verified package retained when a newer fetch, build or activation fails.
Working set	The bounded set of objects selected for propagation at a given cadence; never the entire catalog per visual frame.
Quarantine	Storage and diagnosis state for a source record that failed parsing, validation, identity or propagation checks and is not published.
Pass event	A computed observer-relative interval with rise, maximum elevation and set, based on a specific element set and UTC window.

Appendix G. Orbital Object Ingestion, Identity and Propagation Specification

Normative engineering annex for backend, orbital computation, client platform, UX, QA and operations teams.

Binding status

This appendix is part of OM-CDR-001 v1.1 and is release-controlling. Facts are cited; implementation choices are explicitly classified as design decisions, requirements, thresholds or open items. A team may not replace a specified identity rule, frame convention, package lifecycle or solver without change control.

G.1 Purpose and scope

This annex removes implementation ambiguity from the path between a provider record and a user-visible marker or pass alert. It defines how records enter OrbitMark, how identity is controlled, which data are stored, how positions and passes are computed, how failures are contained, what UX states are required, and what evidence authorizes release.

Class	Statement
F	Providers publish mean-element records; the application computes positions for a requested time. The source is not a live location stream [E-01][E-02][E-22].
D	OMM-compatible JSON is the canonical OrbitMark payload; raw source payload and normalized fields are both retained.
R	Every displayed marker and pass must be traceable to object identity, element-set identity, package version, UTC calculation time and observer context.
R	Only catalogued objects with a valid publishable element set may appear as individually positioned objects. Uncatalogued small debris remains statistical/educational content.

G.2 Responsibility boundary

Owner	Controls	Must not do
Ingest service	Provider access, response handling, parsing, validation, normalization, identity, persistence and package candidate.	Run implicitly in each web process or silently retry blocked providers.
Package publisher	Manifest, checksum, compatibility, atomic current pointer and rollback.	Expose a partial candidate as current.
Orbital worker	OMM-to-SatRec conversion, SGP4/SDP4, coordinate transforms, working-set scheduling, candidate geometry and passes.	Access IndexedDB or perform full-catalog work in the animation frame.
Orientation service	Device/screen frame normalization, confidence state and manual offset.	Claim measured accuracy the platform does not provide.
Presentation layer	Truthful model wording, confidence/freshness, candidates, details and fallbacks.	Describe the camera as detecting or imaging debris.
QA/operations	Reference vectors, fixtures, failure injection, performance, freshness and release evidence.	Approve by visual inspection alone.

G.3 Source policy and precedence

Priority	Source lane	Approved use	Rules
1	CelesTrak GP OMM-compatible JSON	Default public MVP current-element feed.	Conditional fetch; respect update cadence and one-download-per-update behaviour; stop and alert on provider block/error [E-01].
2	Space-Track GP / GP_History	Authenticated secondary validation and controlled history where terms and credentials are approved.	Credentials server-side only; GP is newest set, GP_History is historical [E-02].

Priority	Source lane	Approved use	Rules
3	CelesTrak SupGP	Future explicitly labelled supplemental lane for selected objects.	Never silently replace/mix with GP; source and element-set identity remain visible [E-01].
-	Scraping or unofficial mirrors	Not approved.	Requires new source review and ADR.

Provider conduct requirement

CelesTrak states that it checks for new GP data once every two hours, discourages redundant downloads, and may return blocking responses for excessive requests. OrbitMark therefore performs conditional requests, stores the last provider state, and stops the run on 403/429 or repeated transport errors rather than entering a retry storm [E-01].

G.4 Object identity and eligibility

A catalog number is a source identifier, not OrbitMark's permanent database identity. The database assigns a stable object_key and stores one or more source identities with validity and lineage metadata.

Identity class	MVP treatment	Reason
Public catalog identity	Eligible when source record is valid and not decayed/withdrawn under the publication policy.	Designed for stable catalog use.
Analyst range 80000–89999	Ingest may record for diagnostics, but exclude from public package, Save, Note, Tag and Watch.	Space-Track states analyst numbers are constantly reused for different objects [E-02].
Temporary launch nominal / provisional identity	Exclude from persistent user features in MVP.	Temporary identifiers can change when a permanent catalog identity is assigned.
Unknown/blank catalog ID	Quarantine.	Cannot satisfy traceability or stable user references.
Duplicate source record	Deduplicate by source namespace, catalog ID, epoch and canonical payload checksum.	Prevents duplicate history and non-idempotent runs.

Entity/key	Contract
object_key	UUID generated by OrbitMark; referenced by user data and internal relationships.
source_identity	source_namespace + catalog_id + identity_class + validity interval + lineage_status.
element_set_key	source_namespace + source record ID where available + catalog_id + epoch + canonical checksum.
user reference	References object_key only; user data never binds directly to a reusable analyst number.

G.5 Ingestion state machine

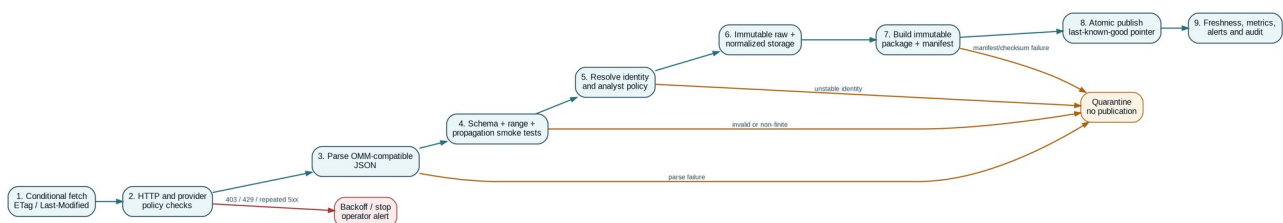


Figure G-1 — Binding ingest, validation and atomic publication pipeline.

44. Acquire a distributed lock and create an ingest_run with source, requested dataset, software version and start time.
45. Issue a conditional request using the last ETag/Last-Modified state where supported. Record HTTP class and relevant headers, never credentials.
46. On 304, finish as no_change. On 403/429, stop immediately, set provider_blocked/backoff, and alert operations. On transient transport/5xx, use capped exponential backoff within the run budget.
47. Parse the complete response into source records. A response body that is HTML/error text or structurally inconsistent with the declared format fails before normalization.

48. Validate required fields, types, UTC epoch, project range guards and safe identifier handling.
49. Create the library SatRec and propagate at the epoch plus controlled offsets; reject non-finite results or a library error.
50. Resolve identity eligibility, deduplicate by element_set_key and write raw plus normalized records immutably in a database transaction.
51. Build a candidate package and manifest from eligible current records. Verify counts, checksum, epoch distribution, compatibility and random propagation samples.
52. Publish atomically by committing the immutable package and moving the current pointer in one controlled operation. A failure preserves the previous pointer.
53. Emit metrics and close the run with success, partial_quarantine, no_change or failed. partial_quarantine requires the M0.5-approved invalid-rate guardrail and no systemic validation pattern.

G.6 OMM normalization contract

Field	Normalized type/unit	OrbitMark rule
NORAD_CAT_ID	String at API boundary; BIGINT/source string in storage	Never assume five digits; reject blank/non-numeric values for current approved sources.
EPOCH	UTC instant	Parse strictly; reject timezone ambiguity; retain original text.
MEAN_MOTION	double, revolutions/day	Required for SGP4 record; finite and positive.
ECCENTRICITY	double, unitless	Required; $0 \leq e < 1$.
INCLINATION	double, degrees	Required; project guard 0..180.
RA_OF_ASC_NODE	double, degrees	Required; normalize contract expects 0..360 from source, no silent unit guessing.
ARG_OF_PERICENTER	double, degrees	Required; 0..360 project guard.
MEAN_ANOMALY	double, degrees	Required; 0..360 project guard.
BSTAR	double, inverse Earth radii in GP convention	Required by the selected library input contract; zero is valid.
MEAN_MOTION_DOT/DDOT	double	Retain when supplied; no product claim is derived directly from these values.
OBJECT_NAME / OBJECT_ID	Unicode text / source identifier	Optional display/metadata; never used as the sole identity key.
REF_FRAME / TIME_SYSTEM / MEAN_ELEMENT_THEORY	controlled metadata	When omitted by provider-specific JSON, the adapter records the documented GP defaults; the raw payload remains unchanged [E-01].

Range-guard classification

The ranges above are project validation guards intended to detect corrupt units and payloads. Except where explicitly defined by a source/standard, they are not represented as universal physical limits. Any guard change requires a fixture and migration review.

G.7 Persistence and immutable history

Store	Mandatory behaviour
raw_source_records	Preserve received record or response fragment, source, fetch time, canonical checksum and parser version. No in-place rewriting.
element_sets	One immutable normalized row per element_set_key, linked to ingest_run and object identity.
object_identities	Separate OrbitMark object_key from source identifiers; retain validity and eligibility decisions.
current_element_selection	A derived pointer/view selecting the approved element set per object/source policy; changing the pointer never deletes history.
quarantine_records	Validation/error codes, source reference, first/last occurrence and disposition.
package_manifests	Immutable version, compatibility, source set, counts, epoch statistics, checksum, size and publication state.

- Database writes are idempotent under the element_set_key and package version.

- History retention is configurable, but the package/version used for any released build and support incident must remain reproducible for the approved retention window.
- Decay/withdrawal is metadata and eligibility state; it does not erase historical records.
- No user note, tag or watch row stores orbital elements or precise observer location.

G.8 Package and manifest contract

Manifest field	Required meaning
version	Globally unique immutable package identifier, for example generated timestamp plus content hash.
format_version	Schema/encoding version interpreted by the client adapter.
required_client_version	Minimum compatible application version.
source_set	Source IDs and dataset classes included; no silent source substitution.
generated_at	UTC package build completion time.
record_count / object_count	Validated eligible counts used for integrity checks.
min_epoch / max_epoch / age buckets	Distribution metadata; not a claim that every element has the same freshness.
checksum / compression / byte_size	Integrity and download contract.
delta_base	Optional prior version from which a delta is valid.
attribution	Provider/source attribution text or identifier controlled by legal review.

```
{
  "version": "2026-06-19T18:00:00Z-7f3c...",
  "format_version": 1,
  "required_client_version": "1.0.0",
  "source_set": [{"id": "celestrak-gp", "dataset": "approved-mvp"}],
  "generated_at": "2026-06-19T18:00:00Z",
  "object_count": 38124,
  "record_count": 38124,
  "min_epoch": "...", "max_epoch": "...",
  "checksum_sha256": "...",
  "compression": "gzip"
}
```

- A package file is written under its immutable version and checksum, then verified from storage before the current pointer changes.
- The client downloads to a staging key, verifies checksum and format compatibility, parses in the worker, and only then marks it active.
- If activation fails, the staged candidate is retained for diagnostics or removed; the last-known-good package remains active.

G.9 Client bootstrap and cache lifecycle

54. Open IndexedDB and read only package metadata plus the last-known-good package version during application bootstrap.
55. Stream/decompress and parse the active package inside the orbital Web Worker. The main thread receives readiness and bounded result messages, not the entire parsed catalog.
56. Expose Home/Manual Sky from cached data when available. If no valid package exists, show a controlled first-load state rather than fabricated positions.
57. Fetch manifest metadata in the background. Download only when version/compatibility and network policy require it.
58. Activate the new package only after checksum, schema and worker parse succeed; then persist the active pointer.
59. Garbage-collect old packages only after the new package has been used successfully and the retention/rollback rule is satisfied.

G.10 Position calculation pipeline

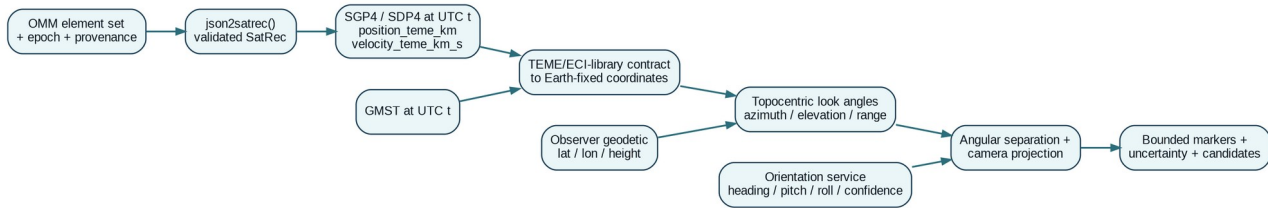


Figure G-2 — Approved position, observer and display pipeline.

Step	Binding computation contract
1. Parse	satrec = json2satrec(normalized_omm). A parse exception or invalid SatRec is a propagation failure, not a zero position [E-03].
2. Propagate	state = propagate(satrec, utc_time). Null/error/non-finite output is classified using the library error and OrbitMark failure code [E-03].
3. Frame/units	Adapter output names are position_teme_km and velocity_teme_km_s. Library ECI labels do not cross the adapter boundary without the explicit convention [E-20][E-22].
4. Earth rotation	gmst = gstime(utc_time); object_ecf_km = eciToEcf(position_teme_km, gmst) under the approved satellite.js adapter contract [E-03].
5. Observer	Observer is geodetic longitude/latitude in radians and height in kilometres relative to the library's Earth model.
6. Topocentric	look = ecfToLookAngles(observer_gd, object_ecf_km), returning azimuth, elevation and range [E-03][E-21].
7. Presentation	Convert radians to display degrees only at the presentation boundary; retain full-precision internal values.

No stored per-second positions

OrbitMark stores element sets and calculates state at a requested UTC time. It does not persist a position for every second. Cached derived samples may be used for performance only when their element-set identity, timestamp and interpolation validity are explicit.

G.11 Time authority and clock-skew control

Rule	Implementation
UTC only	All propagation and pass windows use UTC instants. UI local time is formatting only.
Trusted offset	When online, compare device wall clock with a trusted HTTPS response Date or controlled API time and store an offset estimate with age.
Monotonic intervals	Animation/interpolation durations use a monotonic clock; absolute propagation time uses UTC wall time plus approved offset.
Skew state	Above the project threshold, display "Device time may be inaccurate," lower confidence and avoid precise notification wording.
No silent correction	Do not change the device clock; apply an internal calculation offset only when the method and age are recorded.

G.12 Working-set selection and computation cadence

The full package may be held in worker memory, but propagation is scheduled by priority. Exact counts and cadences are M0.5 profiling outputs stored as configuration, because a universal value would be an unsupported assumption.

Band	Membership	Service objective
A — immediate	Selected object, watched objects with imminent passes, current candidate sheet and explicitly searched IDs.	Highest cadence and no density dropping except on hard safety/performance failure.
B — visible context	Objects recently above/near horizon, within camera/manual viewport margin, or in the	Medium cadence; bounded by density and device budget.

Band	Membership	Service objective
	current overhead list.	
C — background eligibility	Remaining publishable catalog used to discover future Band B objects.	Low-frequency coarse scan or staggered batches; never per visual frame.

60. At package activation, precompute immutable lightweight attributes such as mean motion, object type and element age.
61. At a coarse cadence, process Band C in batches and estimate observer-relative eligibility using the same approved propagation pipeline.
62. Promote candidates near the horizon/viewport and user-priority objects to Band B/A.
63. Propagate Band A and B at configured cadences; publish typed arrays or compact records to the main thread.
64. Interpolate visual marker motion between worker updates when the time gap is inside the approved interpolation limit.
65. Demote objects after hysteresis to prevent rapid band thrashing.
66. Record active counts and compute duration; if a budget is exceeded, reduce Band B density/cadence before affecting Band A.

priority_score =
 selected_bonus + watched_bonus + imminent_pass_bonus +
 viewport_bonus + horizon_proximity_bonus + search_bonus -
 stale_penalty - density_penalty

Note: coefficients are M0.5 configuration values validated by profiling; they are not scientific constants.

G.13 Camera/manual projection and candidate algorithm

67. Represent the object direction as azimuth/elevation and the viewing direction as normalized heading/camera-elevation/roll from the orientation abstraction or Manual Sky.
68. Compute spherical angular separation, not flat screen distance: $\cos(\theta) = \sin(e_1)\sin(e_2) + \cos(e_1)\cos(e_2)\cos(a_1-a_2)$. Clamp the cosine to $[-1,1]$ before $\arccos$.
69. Apply calibrated horizontal/vertical field of view and screen rotation to project eligible directions. The camera FOV value is device/configuration evidence, not assumed from CSS pixels.
70. Construct the candidate radius from the displayed uncertainty region plus the touch-selection tolerance. A marker's visual radius does not define object certainty.
71. Return all candidates inside the angular region. Sort first by angular separation, then explicit user priority and stable tie-breaker; do not silently identify the nearest as certain.
72. When no candidate exists, offer widen region, search, Manual Sky pan and data/confidence help rather than a false match.

G.14 Pass prediction algorithm



Figure G-3 — Deterministic on-device pass solver and notification lifecycle.

For an object, observer and selected element set, define $f(t) = \text{elevation}(t) - \text{horizon_mask}$. The solver uses only the approved propagation pipeline and returns metadata tying every result to `element_set_key` and package version.

73. Estimate orbital period from the SatRec/library fields and choose an adaptive coarse sample step: $\text{clamp}(\text{period_seconds} / 360, 10 \text{ seconds}, 120 \text{ seconds})$. This is a project starting rule subject to G0.5 benchmarking.
74. Sample $f(t)$ across the requested UTC window. A transition $f(t_0) < 0$ to $f(t_1) \geq 0$ brackets rise; the reverse brackets set. Handle a pass already in progress at the window start.
75. Refine each bracket with bisection until interval width is ≤ 1 second or the M0.5-approved tolerance. Bisection is chosen for deterministic bracketing and requires no derivative.
76. Within each rise/set interval, find maximum elevation with a bounded derivative-free scalar search (golden-section or equivalent) and verify the maximum against neighbouring samples.
77. Attach maximum azimuth/range, element epoch/age, sunlit/shadow result when available, observer darkness when implemented, and brightness state (known/unknown).
78. Deduplicate adjacent intervals caused by numerical noise using the approved minimum-gap/tolerance rule and record solver diagnostics.
79. Persist/schedule only the nearest rolling set permitted by the platform. Recompute on app activation, significant location change, package activation, watch change and notification-permission change.
80. Suppress or label an alert when clock state, source age or propagation failure exceeds the approved policy. The app never guarantees OS delivery time.

Pass result field	Meaning
rise_utc / max_utc / set_utc	UTC event instants; local formatting occurs in the UI.

Pass result field	Meaning
max_elevation_deg / azimuth_deg / range_km	Observer-relative computed values at the stated times.
element_set_key / epoch / package_version	Reproducibility and freshness.
solver_version / tolerance_seconds	Algorithm traceability.
sunlight_state / brightness_state / observer_darkness	Independent visibility-related dimensions.
degradation_codes	Stale elements, clock skew, propagation failure or incomplete visibility inputs.

G.15 Visibility and uncertainty domain model

Dimension	Allowed values	UI implication
Geometric state	below_horizon / above_horizon / unknown	Controls whether a direction is geometrically available, not naked-eye visibility.
Sunlight state	sunlit / penumbra_or_uncertain / earth_shadow / unavailable	May support "model indicates sunlit"; never proves visible brightness.
Brightness state	magnitude_known / estimate_only / unknown	Unknown remains unknown; no inferred invisibility.
Observer darkness	daylight / twilight / dark / unavailable	Context only; separate from object brightness.
Element freshness	fresh_by_policy / aging / stale / unknown	Controls wording, alert eligibility and diagnostics.
Orientation confidence	usable / calibrating / low / manual / unavailable	Controls uncertainty region and Camera Sky behaviour.

Permitted summary logic

"May be visible" is allowed only when the approved rule has sufficient geometric, sunlight, observer-light and brightness inputs. Otherwise show the independent facts and "Visibility unknown." Above-horizon alone never maps to visible.

G.16 Failure taxonomy and recovery

Code family	Examples	Recovery / user mapping
ING-HTTP	ING-HTTP-403, -429, -5XX, -TIMEOUT	Stop/backoff/alert operations. User keeps last-known-good; Data Status may show refresh delayed.
ING-PARSE	ING-PARSE-FORMAT, -EMPTY, -HTML	Quarantine run/candidate; no current-pointer change.
ING-SCHEMA	MISSING_FIELD, BAD_TYPE, BAD_EPOCH, RANGE_GUARD	Quarantine record; systemic pattern fails package.
ID	ID-ANALYST-EXCLUDED, ID-TEMP-EXCLUDED, ID-CONFLICT	Do not expose persistent user features; log controlled reason.
PKG	PKG-CHECKSUM, PKG-COUNT, PKG-INCOMPATIBLE, PKG-ACTIVATE	Retain last-known-good and show cached-data state.
ORB	ORB-PARSE, ORB-DECAYED, ORB-NONFINITE, ORB-LIBRARY	Exclude record/result; object detail may show position unavailable.
TIME	TIME-SKEW, TIME-AMBIGUOUS, TIME-SOURCE-STALE	Degrade timing/confidence; suppress high-confidence alert wording.
PASS	PASS-NO-BRACKET, PASS-NO-MAX, PASS-NUMERIC	Do not schedule; retain diagnostics and explain unavailable pass.
SENSOR	SENSOR-DENIED, SENSOR-LOW-CONFIDENCE, SENSOR-UNAVAILABLE	Manual Sky remains complete.

G.17 Verification and numerical reference strategy

Suite	Minimum evidence
Provider fixtures	Captured/redacted approved OMM JSON, 304/no-change, 403/429,

Suite	Minimum evidence
	HTML error body, truncation and malformed records.
Schema/property	Required fields, type coercion rejection, large IDs, UTC parsing, range guards and canonical checksum stability.
SGP4 reference	Vallado verification vectors and an independent implementation such as python-sgp4; compare position/velocity and observer look angles [E-20].
Coordinate fixtures	Known UTC, observer and object states covering quadrant boundaries, horizon crossings, longitude wrapping and high latitude [E-21].
Identity fixtures	Public identity, analyst number reuse, temporary-to-permanent migration and conflicting source metadata.
Pass fixtures	No pass, pass in progress, grazing horizon, multiple passes, deep-space object, stale element and clock skew.
Package failure	Interrupted write, wrong count/checksum, incompatible format, parse crash and rollback.
Performance	Supported devices, active counts, worker compute duration, memory, main-thread long tasks, battery/thermal sampling and background resume.

- Golden fixtures are version-controlled and name the source/license status, expected implementation, tolerance and reason for inclusion.
- A test passes only when both numerical output and classification/failure metadata match.
- No single third-party library validates itself: OrbitMark compares against an independent implementation or published reference outputs.
- The G0.5 evidence pack contains machine-readable results, commit/package versions and a human-readable variance summary.

G.18 Observability and operational controls

Signal	Required dimensions
ingest_run_total / duration	source, dataset, result, HTTP class, parser version.
source_freshness	source, latest provider/fetch time, current package age, oldest/median/newest element-age buckets.
record_validation	validation code, count, source, systemic flag; never raw payload in telemetry.
package_publish	version, count, size, checksum result, required client, duration, rollback flag.
client_activation	version, success/failure code, parse duration, device tier; no location.
orbital_compute	worker duration buckets, active bands/counts, failure-code buckets, package version.
pass_compute	object type, window bucket, event count, solver duration, degradation-code bucket; no coordinates.

- Operations has a runbook for provider block, stale source, invalid-rate spike, package publish failure, client incompatibility and numerical regression.
- Alerts distinguish “provider not updated” from “OrbitMark ingest failed.”
- Raw precise user location, free-form notes/tags, camera frames and headings remain prohibited telemetry.

G.19 Security, privacy and provider controls

- Space-Track credentials and provider session tokens exist only in the server secret store and dedicated ingest process.
- Provider payloads are treated as untrusted input: bounded size, strict content type/format, parser limits, schema validation and no dynamic code execution.
- Package checksum protects integrity; transport uses HTTPS. A future signature scheme may be added by ADR if threat modelling requires it.
- Observer coordinates used for sky/pass calculation remain local by default. Diagnostics use coarse/non-sensitive categories.
- Source terms, attribution, caching and redistribution require the existing legal gate before public production.

G.20 UX/UI implementation obligations

Surface	Required data-engine states
S-01 Launch	Loading cached manifest, activating package, last-known-good fallback, incompatible package and first-load unavailable.
S-03 Tonight	Package/source age summary, observer/time state and honest empty state; no "live" label for computed data.
S-05/S-06 Sky	Modelled-time indicator, source/freshness access, orientation confidence, clock-skew warning, candidate region and position-unavailable state.
S-07 Candidate list	Angular candidate region, count, separation ordering, object type/identity limits and no silent nearest-object certainty.
S-08 Object detail	Object key-facing UI uses catalog ID/source, element epoch, package version, calculated-for time, geometric/visibility fields and degradation reasons.
S-11/S-12 Passes	Element/source age, solver result, local-time formatting, alert permission/delivery caveat, recompute state and stale/skew suppression.
S-15 Data Status	Source lane, last success, active package, element-age distribution, cached fallback, provider delay, quarantine/systemic error and support export.
S-18 Error/offline	Specific recoverable message mapped from controlled code; preserve useful cached/manual functionality.

Claude Design requirement

Claude Design must produce complete visual and accessibility states for package activation, cached fallback, stale/aging elements, provider delay, clock skew, position unavailable, excluded/unstable identity, pass unavailable and diagnostic export. A generic red error banner is not an acceptable substitute.

G.21 Team ownership and integration sequence

Order	Team/owner	Deliverable	Downstream dependency
1	Orbit/backend	Canonical schemas, source adapter, identity rules, fixtures and ingest state machine.	Package publisher, QA.
2	Platform/backend	Immutable package/manifest, API/status and operations runbook.	Client cache/worker.
3	Orbital client	SatRec adapter, explicit frames/units, working-set worker and diagnostics.	Sky/manual UI.
4	Pass/mobile	Pass solver, clock/freshness policy and local notification lifecycle.	Passes screens.
5	UX/content	All Appendix G state/copy contracts and accessible fallbacks.	Frontend implementation and tests.
6	QA/security/operations	Independent vectors, failure injection, performance, privacy/provider review and evidence pack.	G0.5 approval.

G.22 Gate G0.5 — development authorization checklist

Gate item	Pass evidence
Source adapter	Provider-policy tests, conditional fetch, block/backoff and no credential leakage.
Identity	Analyst/temporary exclusion and reuse fixtures; stable object_key relationships.
Persistence	Immutable raw/normalized sets, idempotency and no history overwrite.
Package	Manifest, checksum, atomic pointer, interrupted-write and client rollback tests.

Gate item	Pass evidence
Numerics	Reference-vector parity for OMM parsing, SGP4, coordinate transforms and errors.
Working set	Profile proving no full-catalog per-frame work and acceptable supported-device budgets.
Pass solver	Rise/max/set reference parity, stale/skew degradation and reschedule tests.
UX states	Reviewed S-01/S-03/S-05/S-08/S-11/S-15/S-18 designs for every required engine state.
Operations	Metrics, alerts, runbook and diagnostic export.
Approval	Orbit owner, architecture, QA and Product Owner sign the evidence pack.

Stop-work rule

Broad screen implementation may prototype against fixtures, but Camera Sky, Passes and public data refresh are not production-authorized until G0.5 passes. A failed gate is not waived through visual polish or schedule pressure.

G.23 Definition of Done for orbital-data features

- Requirement/ADR IDs are linked in code, tests and pull-request evidence.
- Inputs, units, frames, time system and identity class are explicit at module boundaries.
- Happy path, stale/offline, provider error, malformed record, clock skew, propagation failure and rollback are implemented and tested.
- No raw location, credentials, notes or camera data appear in logs/analytics.
- Performance results exist for supported device tiers and preserve Manual Sky.
- UX copy passes the claims registry and accessibility states pass screen-reader/large-text review.
- Operations can diagnose the package/source/solver version from non-sensitive support information.
- The release evidence pack can reproduce the displayed position/pass from fixture inputs.

G.24 Controlled open items

ID	Open item	Resolution rule
G-O-01	Element-age guardrails by orbit/object class.	Set from validation and user-risk evidence at G0.5/G1; do not invent one global scientific limit.
G-O-02	Exact working-set counts/cadences and interpolation limit.	Set by profiling on supported device tiers and stored in versioned configuration.
G-O-03	Whether SupGP is introduced for selected objects.	Requires provider/legal review, explicit UI source label and new ADR/source-precedence tests.
G-O-04	Whether analyst identities can later be lineage-mapped.	Requires durable mapping evidence, migration tests and privacy/product review before user persistence.
G-O-05	Package cryptographic signature beyond checksum.	Threat-model decision before public scale; checksum remains mandatory.

Appendix H. Autonomous Development Operating Model

Binding execution authority for unattended development from repository bootstrap to a production-candidate staging build. This appendix does not authorize production release or external/legal sign-off.

H.1 Approved operating defaults

Area	Controlled default
Project state	Greenfield monorepo; no legacy-code compatibility assumption.
Client	React + TypeScript + Vite PWA, followed by Capacitor iOS/Android packaging.
Services	Flask API, PostgreSQL, dedicated ingest worker and Railway staging.
Developer environment	WSL2, Claude Code CLI, GitHub, PyCharm and Git worktrees.
Automation authority	Dependencies, source edits, disposable/staging migrations, tests, browsers, screenshots, commits, feature pushes, PRs and staging deployments.
Protected boundary	No main merge, production deployment/destructive data, billing, secret disclosure, provider/legal acceptance or gate bypass.
Devices	Browser/simulator evidence first; physical iPhone/Android gate remains externally blocked until devices are named.
Design output	Figma-ready specification, tokens, SVG/PNG, Storybook and implementable React components.
Blocker behavior	Continue with fixtures/mocks and safe conservative defaults; record blockers and stop only the blocked irreversible capability.

H.2 Autonomous control loop

1. Read RUN_STATE.yaml, WORK_QUEUE.yaml, controlling CDR clauses, blockers and previous evidence.
2. Select one highest-priority READY work package with satisfied dependencies and no file-ownership conflict.
3. Create a feature branch/worktree and a work-package contract binding requirements, files, tests, screenshots, evidence, rollback and prohibited actions.
4. Run the planning Council quorum and resolve every STOP/CRITICAL issue before editing.
5. Implement the smallest coherent vertical slice, using fixtures when external dependencies are unavailable.
6. Run layered verification, launch and drive the real app, capture required screenshots and generate an evidence manifest.
7. Run independent Council review; the implementation agent cannot approve its own work.
8. Remediate and re-run until no unresolved STOP/CRITICAL/MAJOR finding remains.
9. Commit, push the feature branch, open/update a PR, update traceability/state and evaluate the work-package/phase gate.
10. Continue automatically to the next READY package, or return a durable blocker/failed-safe state.

Autonomous delivery control loop

One work package at a time; deterministic gates and independent council review before promotion.

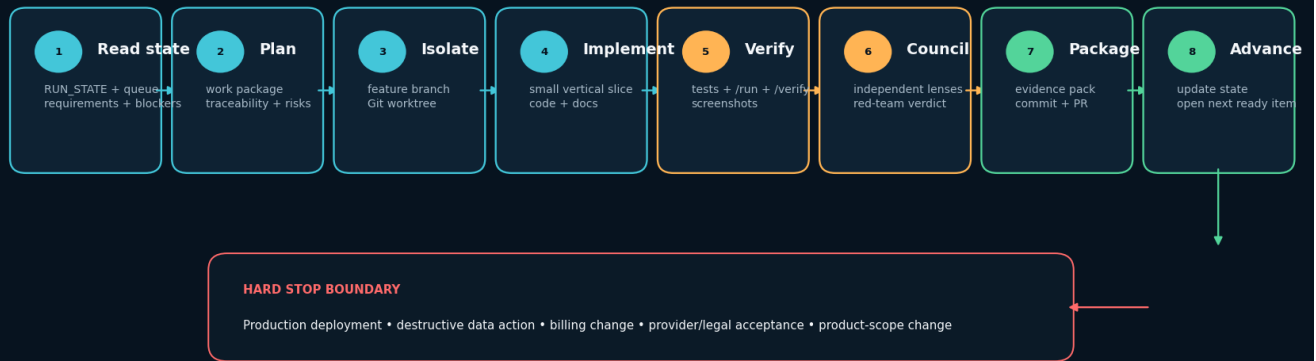


Figure H-1 - Autonomous delivery control loop. Evidence and Council review precede every gate transition.

H.3 State and work-package model

State	Meaning	Allowed next states
BACKLOG	Defined but dependencies/gate not satisfied.	READY
READY	Eligible for autonomous selection.	PLANNING
PLANNING	Contract and planning Council in progress.	IMPLEMENTING, BLOCKED_*
IMPLEMENTING	Code/docs being changed in isolated branch/worktree.	VERIFYING, BLOCKED_*
VERIFYING	Layered tests/running-app/screenshots/evidence.	COUNCIL_REVIEW, REMEDIATION
COUNCIL_REVIEW	Independent specialist review.	REMEDIATION, EVIDENCE_COMPLETE
REMEDIATION	Findings are fixed and reverified.	VERIFYING, COUNCIL_REVIEW
EVIDENCE_COMPLETE	Required evidence/traceability complete.	DONE, BLOCKED_*
DONE	Work package and its exit checks passed.	None
BLOCKED_EXTERNAL	Credential, provider/legal, device or Product Owner authority required.	READY after evidence/authority
BLOCKED_TECHNICAL	Retry budget exhausted with reproducible failure.	READY after new diagnosis/decision
FAILED_SAFE	No integration; branch/evidence preserved.	READY after incident resolution

H.4 Phase and gate sequence

Phase	Controlled scope	Exit gate
P0	Governance, repository controls, Skills/agents/hooks, CI and state.	G-P0
P1	PWA/API/database/Storybook clean foundation.	G-P1
P2	OMM ingestion, identity, immutable storage, quarantine and package	G-P2

Phase	Controlled scope	Exit gate
	lifecycle.	
P3	Orbital engine, explicit frames/units, golden vectors, working set and client activation.	G0.5
P4	UX design system, S-01..S-18 contracts/states and Storybook baselines.	G-P4
P5	Tonight, Manual Sky, candidates, object detail and local catalog.	M1 manual core
P6	Camera/sensors/calibration/uncertainty with fallback.	M1 Camera conditional
P7	Local passes, watch recomputation and notifications.	M2
P8	Data Status, Learn, settings/privacy/a11y and trust.	M3
P9	Security, accessibility, performance, resilience and localization hardening.	Beta
P10	Railway staging beta, migration/rollback and evidence closure.	Staging production candidate

H.5 Gate decisions and retry policy

- PASS continues automatically; PASS_WITH_ADVISORY continues and creates backlog items.
- BLOCK triggers automatic remediation within the work-package retry budget.
- EXTERNAL_BLOCK keeps the affected capability/gate closed but does not stop independent work.
- STOP_SHIP freezes integration and creates a high-severity incident/evidence record.
- Three repetitions of the same deterministic failure signature trigger debugger and red-team review, then BLOCKED_TECHNICAL if unresolved.
- Provider 403/429 stops the ingest run immediately; no aggressive retries; last-known-good data remains active.
- Security, privacy, claims and data-loss critical findings cannot be waived by the autonomous system.

H.6 Repository and branch authority

- Use main as protected production history; unattended processes cannot merge or deploy it.
- Use develop as staging integration; auto-merge is permitted only after all mandatory checks and only if repository policy enables it.
- Use wp/<phase>-<id>-<slug> branches and optional Git worktrees, with one writer per worktree.
- Every stateful/database change includes a versioned migration and rollback/recovery evidence.
- Conversational memory is never the run state; durable YAML, work-package files, commits and evidence control resumption.

H.7 Required autonomous artifacts

Artifact	Purpose
CLAUDE.md	Concise project facts, authority and safety boundaries.
.claude/skills/**/SKILL.md	Repeatable project procedures.
.claude/agents/*.md	Independent specialist Council contexts.
.claude/settings.json	Shared permissions, sandbox and deterministic hooks.
RUN_STATE.yaml / WORK_QUEUE.yaml	Resumable state machine and dependency graph.
DOCS/PROJECT_SKILLS.md	Generated human-readable Skills registry; CI-enforced.
DOCS/12_BLOCKERS.md	External/technical blocks and safe continuation.
artifacts/evidence/**/manifest.json	Hashed evidence tied to commit/requirements/gate.
COUNCIL_REVIEW.md / findings JSON	Independent review record.
Gate record	Only authority for phase progression.

H.8 Autonomous completion boundary

Autonomous delivery ends with a production-candidate staging build and complete residual-risk/external-block list. Production merge/deployment, provider/legal acceptance and mandatory external professional approvals remain human-controlled.

Appendix I. Claude Skills, Agents and Council Strategy

Binding strategy for using Claude Code capabilities without relying on unstructured prompting or experimental controls as the sole quality mechanism.

I.1 Operating split

Mechanism	Use	Control rule
CLAUDE.md	Always-loaded concise facts, commands, architecture and boundaries.	Keep concise; procedures belong in Skills.
Project Skills	Repeatable multi-step workflows under .claude/skills.	Explicitly invoked for critical gates; evaluated and registered.
Subagents	Isolated specialist contexts and independent Council review.	Implementation agent cannot self-approve; tools/permissions restricted.
Command hooks	Deterministic pre/post/stop enforcement.	Production controls use shell/CI, not model judgment.
CI	Reproducible checks and evidence retention.	Branch protection and required checks remain authoritative.
Bundled run/verify skills	Launch and inspect the actual app.	Record a project run recipe for multi-service launch.
Agent teams/dynamic/agent hooks	Optional experiments for parallel audits.	Never the sole gate; maintain deterministic fallback.

I.2 Mandatory Skills portfolio and registry

Skill	Primary authority/output
orbitmark-bootstrap	Initialize or validate the OrbitMark repository, environment, autonomous controls, state, work queue, and generated skills registry.
orbitmark-autopilot	Execute one bounded autonomous OrbitMark work-package cycle from queue selection through planning, implementation, verification, council review, evidence, PR and state update.
orbitmark-plan-work-package	Create a controlled work-package plan binding requirements, files, tests, screenshots, risks, council roles, evidence and rollback before implementation.
orbitmark-requirements-traceability	Maintain bidirectional traceability among CDR requirements, decisions, work packages, code, tests, screens, evidence and release gates.
orbitmark-orbital-data	Implement or review OrbitMark orbital ingestion, identity, immutable packages, SGP4/SDP4 propagation, frames, look angles, working sets and pass calculations under Appendix G.
orbitmark-ux-design-authority	Design OrbitMark screens, states, components and visual system for Claude Design while enforcing truth, curation, accessibility, Manual Sky parity and data-engine states.
orbitmark-frontend-implementation	Implement approved OrbitMark React/TypeScript UI contracts as semantic, responsive, accessible components with Storybook and browser tests.
orbitmark-test-and-evidence	Run the required OrbitMark test layers and create an immutable, hashed evidence pack tied to requirements and a commit.
orbitmark-screenshot-matrix	Capture deterministic OrbitMark screen and component screenshots across required states, viewports, text scales

Skill	Primary authority/output
	and themes, and compare them to approved baselines.
orbitmark-accessibility-gate	Evaluate OrbitMark critical tasks and UI states for WCAG 2.
orbitmark-security-privacy-gate	Review OrbitMark code, architecture and evidence for secrets, dependency risk, auth/API abuse, local-first location, data minimization and release security.
orbitmark-claims-gate	Scan OrbitMark UI, documentation, tests and marketing copy for prohibited or unsupported detection, accuracy, freshness, visibility, debris and notification claims.
orbitmark-council-review	Run the appropriate independent OrbitMark council quorum and consolidate structured findings without allowing implementers to approve their own work.
orbitmark-release-gate	Evaluate an OrbitMark work-package, phase or release gate strictly from required evidence and issue a machine-readable PASS/BLOCK/EXTERNAL_BLOCK/STOP_SHIP record.
orbitmark-docs-sync	Synchronize OrbitMark requirements, ADRs, contracts, diagrams, APIs, screenshots and runbooks after implementation changes.

R-I-001: At bootstrap and after every Skills change, Claude shall regenerate DOCS/PROJECT_SKILLS.md from .claude/skills/**/SKILL.md. CI shall fail on missing, duplicate or orphan registry entries.

I.3 Skill engineering and evaluation

- 1. Use a narrow action-oriented name and trigger description.
- 2. Declare controlling CDR clauses, inputs, outputs, preconditions, prohibited operations and completion evidence.
- 3. Use deterministic scripts for validation, destructive-command guards, registry generation and gate decisions.
- 4. Include positive, edge/failure and prohibited-operation fixtures.
- 5. Evaluate material changes; record expected vs. observed artifacts and trigger behavior.
- 6. Disable and mark BLOCKED any skill that over-triggers, modifies protected files, weakens gates or requests unsafe tools.
- 7. Community/external skills are advisory and require source/license/security/evaluation review before activation.

I.4 Council architecture

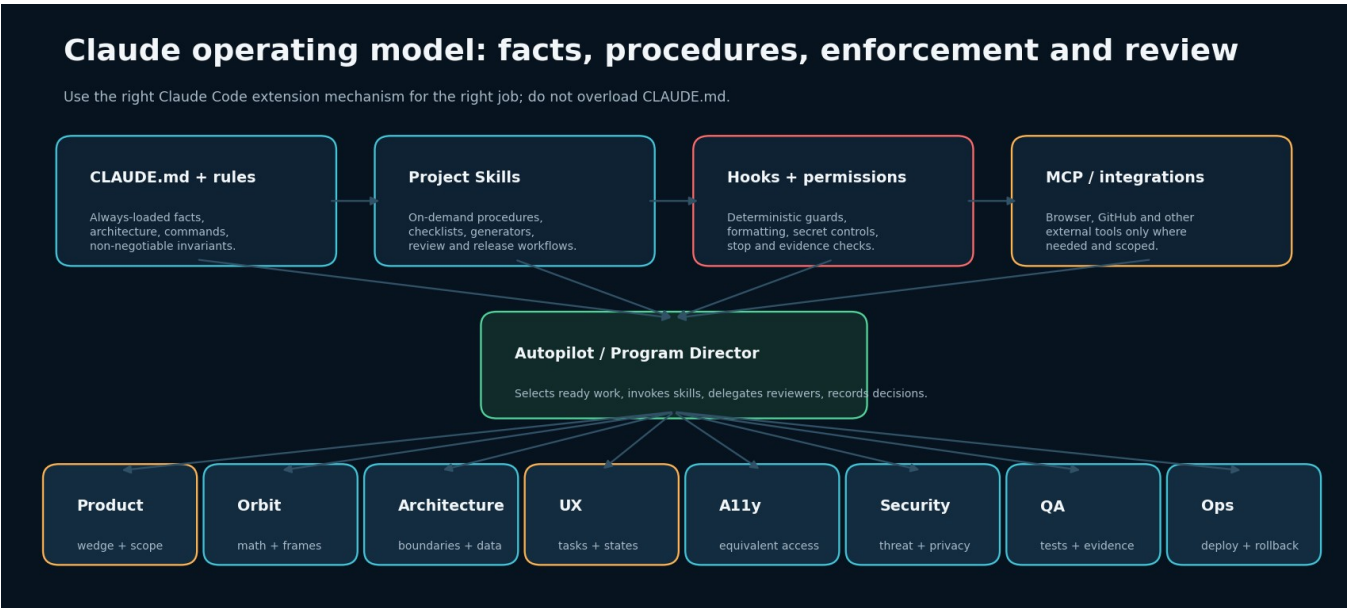


Figure I-1 - Skills execute controlled procedures; specialist subagents challenge work; deterministic hooks and CI enforce the boundary.

Council role	Mandatory review focus
Program Director	Coordinate OrbitMark autonomous delivery, work queue,

Council role	Mandatory review focus
	gates, council and evidence. Use for planning and phase control.
Product Strategy Reviewer	Protect the OrbitMark annotation/curation wedge, MVP boundary and business-plan evidence. Use for product and scope review.
Orbital Dynamics Reviewer	Independently review OMM, SGP4/SDP4, frames, UTC, look angles, uncertainty, working sets and pass solvers.
Data Platform Architect	Review ingestion, identity, immutable persistence, packages, APIs, migrations, rollback and observability.
Mobile Ar Architect	Review PWA/Capacitor, sensors, orientation, camera projection, permissions, notifications and device performance.
Ux Design Director	Review information architecture, visual direction, screen/state completeness and implementation handoff.
Accessibility Auditor	Audit non-camera parity, semantics, keyboard, screen reader, text scale, contrast and reduced motion.
Security Privacy Reviewer	Audit threats, secrets, dependencies, authorization, data minimization and precise-location handling.
Qa Reliability Lead	Challenge test coverage, determinism, failure injection, visual evidence, flakiness and reproducibility.
Devops Release Manager	Review CI/CD, environments, migrations, rollback, staging, observability and release evidence.
Commercial Reality Reviewer	Review cost, complexity and commercial relevance against the business plan without changing product authority.
Red Team Adversary	Attempt to falsify completion, identify unsafe assumptions, reproduce failures and challenge claims/evidence.

I.5 Quorums and severity

- Planning: Program Director, owning domain reviewer, QA and one adversarial reviewer.
- Design: UX, Accessibility, Product/Claims and Frontend/Mobile.
- G0.5: Orbital, Data Platform, QA, Security/Privacy and Program Director.
- Phase gate: Program Director, QA, impacted specialists and Red Team.
- STOP/CRITICAL/MAJOR findings must be remediated before work-package completion; missing evidence cannot be downgraded by opinion.

I.6 Permissions and hooks

- Project settings default to acceptEdits within the sandbox; risky Git push, staging deploy and migration commands require explicit permission/rules.
- Deny main/master pushes, destructive Git cleanup, root deletion, production commands and Railway deletion.
- PreToolUse command hook inspects Bash commands and returns a blocking exit for protected patterns.
- PostToolUse runs fast deterministic checks after edits; Stop verifies evidence when state claims completion.
- Agent/model hooks, if piloted, remain advisory because command hooks are the production-stable enforcement path.

I.7 Source basis

Primary source	Applied conclusion
Claude Code Skills documentation	Project procedures are packaged in .claude/skills and load on use; run/verify workflows are recorded as project skills.
Claude Code Subagents documentation	Specialist agents use isolated contexts, focused prompts and restricted tools/permissions.
Claude Code Settings/Memory documentation	Project settings/hooks are shared; CLAUDE.md carries concise startup context, not enforcement.
Claude Code Hooks documentation	Command hooks provide deterministic lifecycle control; agent hooks are experimental.

Primary source	Applied conclusion
Claude Code CLI/Common Workflows	Programmatic print mode, budgets/turn limits, branches and worktrees support bounded automation.
Anthropic official Skills repository	Official frontend-design/skill-creator material may inform design and evaluation but remains subordinate to the CDR.

Appendix J. Claude Design Production and Automated Evidence

Binding production package for Claude Design, frontend implementation, screenshots, visual regression and release evidence.

J.1 Design production scope

- Experience strategy, JTBD, critical journeys, permission/trust model and accessibility risk analysis.
- Information architecture and deep-link/state transition model.
- S-01 through S-18 screen contracts with full default/degraded/error/accessibility states.
- Design tokens, typography, spacing/grid, marker grammar, icons, motion, focus and data-density rules.
- Figma-ready frames/specification, SVG/PNG assets, Storybook stories and React-ready component contracts.
- Responsive/mobile/tablet/desktop designs, dynamic text, reduced motion, keyboard and screen-reader behavior.
- Implementation handoff, copy/claims deck, design decisions, known limitations and evidence matrix.

J.2 Visual and interaction authority

Principle	Binding consequence
Calm night instrument	Deep navy surfaces, high contrast, restrained cyan/amber/status colors; avoid generic neon sci-fi and HUD clutter.
Curation over spectacle	Catalog/save/note/tag/watch/pass hierarchy remains primary.
Visible truth	Modelled-for time, source/package/element age, clock/sensor confidence and visibility unknowns are prominent.
Permission-late	Show value first; request each permission only after user action and a primer; complete fallback remains available.
Candidate certainty	Tap returns all candidates within angular/uncertainty region; never silently claim one exact object.
Accessible parity	No core task depends on camera, motion, color, drag, fine pointing, canvas or animation.
Signature motif	Dashed uncertainty ring communicates bounded, model-based direction.

J.3 Screen/state evidence matrix

- Capture each material screen/state at 390x844 and 1280x800; critical flows add 360x800, 430x932, 768x1024 and 1440x900.
- Capture 100%, 150% and 200% text where supported, reduced motion, offline, keyboard/focus and pseudo-localization.
- Platform-native camera/sensor/notification permission states are simulator/physical-device evidence, not browser baselines.
- File name: <screen-id>__<state>__<viewport>__<theme>__<text-scale>__<locale>__<commit>.png.
- Every screenshot has a JSON sidecar with route, fixture, package/element, browser/device/OS, pixel ratio, timestamp, commit, test and requirement IDs.

J.4 Automated verification pipeline

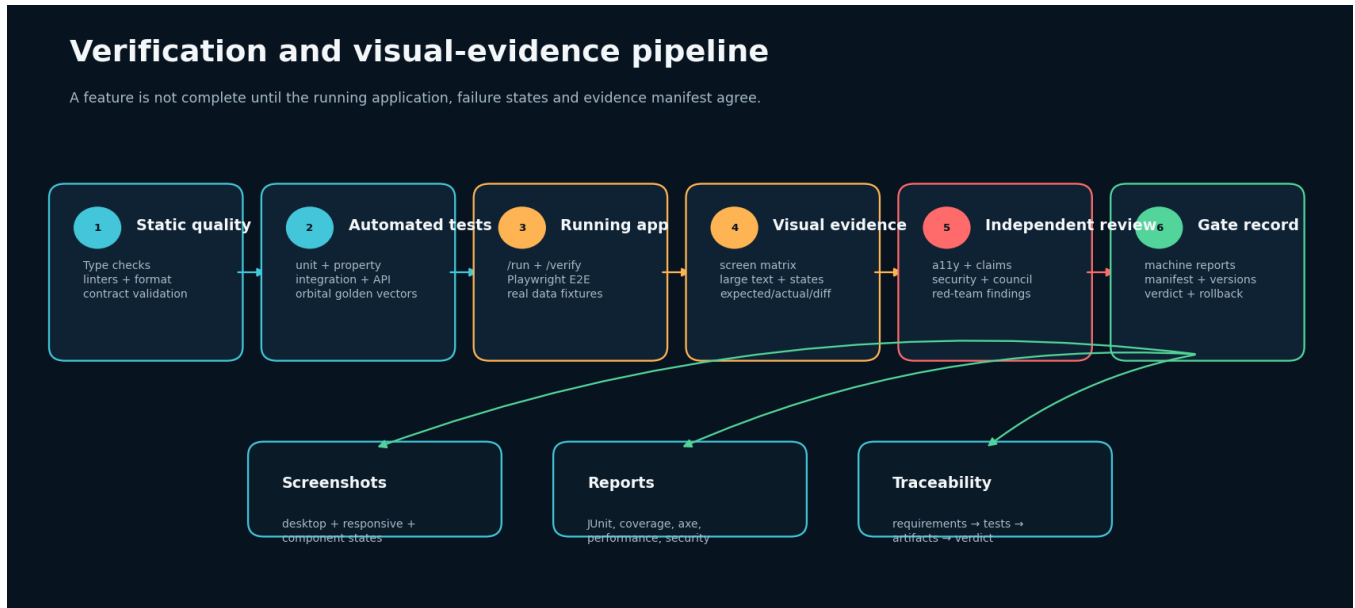


Figure J-1 - Layered checks feed a hashed evidence pack and independent Council gate.

Layer	Required tool/evidence
Static	Type/lint/format/schema plus secret/dependency scans.
Unit/property	Vitest/Pytest, orbital/time/identity edge fixtures.
Contract/integration	OpenAPI/JSON Schema, migrations, ingest/package/client rollback.
Component	Storybook render/interaction/a11y plus Vitest Browser Mode.
E2E	Playwright critical flows across supported browsers.
Visual	Playwright screenshots and Storybook baselines with actual/expected/diff.
Accessibility	axe, ARIA snapshots, keyboard/text scale/contrast plus manual AT checklist.
Performance/resilience	Worker/main-thread profiles, offline/network/provider/package failure injection.
Device	Named iPhone/Android camera/sensor/notification matrix before relevant gates.
Evidence	Environment, commands, logs, reports, traces, screenshots, hashes, traceability, Council and gate record.

J.5 Baseline and screenshot governance

- Freeze clock, location, package, element set, locale, random seed and animation state for deterministic captures.
- Never blanket-accept visual diffs. An approved visual-change record links the changed requirement/design decision and Council review.
- Screenshots supplement behavioral assertions and cannot prove interaction, accessibility or numerical correctness alone.
- Automated axe findings supplement and do not replace manual assistive-technology review.
- Simulator evidence supplements and does not replace required physical-device evidence.

J.6 Design acceptance scorecard

Dimension	Weight	Blocking rule
Product wedge/task clarity	20	Must make curation loop primary.
Truth/uncertainty/data state	20	Any unsupported certainty is STOP.
Accessibility/non-camera parity	20	Any inaccessible critical task is STOP.
Visual hierarchy/distinctiveness	15	Generic/low-legibility direction fails

Dimension	Weight	Blocking rule
		review.
State completeness/recovery	10	Missing material failure state blocks.
Responsive/platform fit	10	Critical clipping/platform mismatch blocks.
Performance/implementation feasibility	5	Unbounded or main-thread-heavy design blocks.

Threshold: at least 85/100 and no truth, accessibility, privacy, claims or core-task critical defect.

J.7 Release evidence record

- Each work package stores an immutable evidence directory with manifest and SHA-256 hashes.
 - The manifest binds commit, environment, requirements, commands, outputs, Council findings and gate decision.
 - Gate progression is denied when evidence is missing, stale, unhashed or produced from a different commit.
 - Production remains human-controlled even when every machine-verifiable item passes.
- Revision v1.2 note: Appendices H-J establish the autonomous development operating model, Skills/Council governance and Claude Design/evidence production package. They do not alter the core product truth, orbital-data or external sign-off boundaries in v1.1.*